

Automated Vital Sign Monitoring in Non-ICU Settings with Wrist-worn Wearable

Péter Lakatos

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Supervisors: Dr. Yogesh Todarwal, Miklós Knébel (Entremo)

Examiner: Saikat Chatterjee

School of Electrical Engineering and Computer Science

Host company: Entremo

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Abstract

Continuous monitoring of vital signs is an essential element of safe patient care, yet current hospital practice still relies heavily on intermittent manual measurement or wired monitoring systems that limit patient mobility. This thesis addresses the problem of whether a wrist-worn, wireless device can reliably monitor heart rate, oxygen saturation, and skin temperature in a way that optimizes nursing workload while maintaining patient comfort and safety. Furthermore, another focus area of the thesis was to explore stakeholder needs and implement them into the user experience design of the prototype.

While wearable devices exist for fitness tracking, they generally do not meet the accuracy, reliability, or workflow integration requirements of clinical settings. Hospitals need solutions that are accurate, effortless for staff to operate, and robust under real-world conditions. Developing such a device requires not only technical innovation in sensing and signal processing but also an understanding of interaction design.

The developed Entremo prototype device combines optical sensors and temperature measurement in a compact, wireless wristband with a screen-free, button-free interface. Key design choices included multi-day battery life with magnetic charging, wipe-down sanitization, and robust signal processing with filtering and signal quality gating. These decisions were guided by interviews with nurses, doctors, and patients, ensuring alignment with practical requirements. Validation experiments were performed on a small sample of healthy volunteers, comparing Entremo's outputs to clinical reference devices in various real-life circumstances.

The results showed that heart rate could be measured within ±3 beats per minute of a reference chest strap, demonstrating feasibility for clinical use. Oxygen saturation measurements were promising but limited by the absence of a dedicated desaturation calibration study. Skin temperature readings were stable enough for trend detection but did not yet meet clinical thermometer standards. These findings indicate that a wireless wrist-worn device can deliver sufficiently accurate heart rate monitoring, while oxygen saturation and temperature measurement require further refinement.

The thesis contributes a proof-of-concept design that demonstrates feasibility and identifies critical development paths for future work, such as accelerometer sensor fusion or fall detection. These findings are aligned with the current industry sentiment that wearable medical devices are the future of continuous patient monitoring, resulting in higher quality of care, lower costs and more satisfied patients.

Keywords

wearable devices, patient monitoring, heart rate, oxygen saturation, skin temperature, photoplethysmorgraphy

Sammanfattning

Kontinuerlig övervakning av vitala funktioner är en viktig del av säker patientvård, men dagens sjukhusrutiner är fortfarande i hög grad beroende av manuella mätningar eller trådbundna övervakningssystem som begränsar patientens rörlighet. Denna avhandling behandlar frågan om en trådlös enhet som bärs på handleden på ett tillförlitligt sätt kan övervaka hjärtfrekvens, syremättnad och hudtemperatur på ett sätt som optimerar vårdpersonalens arbetsbelastning samtidigt som patientens komfort och säkerhet upprätthålls. Ett annat fokusområde för avhandlingen var att utforska intressenternas behov och implementera dem i prototypens användarupplevelsedesign.

Det finns visserligen bärbara enheter för fitnessspårning, men de uppfyller i allmänhet inte kraven på noggrannhet, tillförlitlighet eller arbetsflödesintegration i kliniska miljöer. Sjukhus behöver lösningar som är noggranna, enkla för personalen att använda och robusta under verkliga förhållanden. För att utveckla en sådan enhet krävs inte bara teknisk innovation inom sensorer och signalbehandling, utan också en förståelse för interaktionsdesign.

Den utvecklade Entremo-prototypen kombinerar optiska sensorer och temperaturmätning i ett kompakt, trådlöst armband med ett skärmlöst och knappfritt gränssnitt. Viktiga designval inkluderade batteritid på flera dagar med magnetisk laddning, avtorkningsbar desinficering och robust signalbehandling med filtrering och signalkvalitetsgranskning. Dessa beslut baserades på intervjuer med sjuksköterskor, läkare och patienter för att säkerställa att de överensstämde med praktiska krav. Valideringsexperiment utfördes på ett litet urval av friska frivilliga, där Entremos resultat jämfördes med kliniska referensenheter i olika verkliga situationer.

Resultaten visade att hjärtfrekvensen kunde mätas med en noggrannhet på ±3 slag per minut jämfört med en referensbröstrem, vilket visar att den är lämplig för klinisk användning. Mätningarna av syremättnad var lovande men begränsades av avsaknaden av en särskild kalibreringsstudie av desaturation. Hudtemperaturmätningarna var tillräckligt stabila för trenddetektering, men uppfyllde ännu inte kliniska termometerstandarder. Dessa resultat indikerar att en trådlös handledsenhet kan leverera tillräckligt noggrann hjärtfrekvensövervakning, medan syremättnad och temperaturmätning kräver ytterligare förfining.

Avhandlingen bidrar med en proof-of-concept-design som visar på genomförbarheten och identifierar viktiga utvecklingsvägar för framtida arbete, såsom accelerometersensorfusion eller fallavkänning. Dessa resultat stämmer överens med den rådande uppfattningen inom branschen att bärbara medicintekniska produkter är framtiden för kontinuerlig patientövervakning, vilket resulterar i högre vårdkvalitet, lägre kostnader och nöjdare patienter.

Nyckelord

bärbara enheter, patientövervakning, hjärtfrekvens, syremättnad, hudtemperatur, fotopletysmografi

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List of acronyms and abbreviations

AAC	Alternating current amplitude
AC	Alternating current
ADC	Direct current offset
AES	Advanced encryption standard
AF	Atrial fibrillation
AFE	Analog front end
BP	Blood pressure
BLE	Bluetooth Low Energy
BPM	Beats-per-minute
CBOR	Concise Binary Object Representation
CE	Conformite Europenne
DC	Direct current
DRF	Django Rest Framework
EU	European Union
FDA	Food and Drug Administration
HR	Heart rate
HRV	Heart rate variability
ICU	Intensive care unit
IR	Infrared
LED	Light-emitting diode
LoA	Limits of Agreement
MAE	Mean absolute error
MDR	Medical Device Regulation
ML	Machine learning
MRI	Magnetic resonance imaging
NBA	National Basketball Association
NCIT	Non-contact infrared thermometer
NTC	Negative temperature coefficient
PCB	Printed circuit board
PCM	Protection circuit module
PD	Photodiode
PPG	Photoplethysmography
PSD	Power spectral density
RMSE	Root mean squared error
RR	Respiratory rate
SD	Standard deviation
SNR	Signal-to-noise ratio
SQI	Signal quality indicator
SpO_2	Oxygen saturation
SR	Sampling rate
UDP	User datagram protocol
VPN	Virtual private network

1 Introduction

This chapter describes the specific problem that this thesis addresses, the context of the problem, the goals of this thesis project, and outlines the structure of the thesis.

1.1 Background

The healthcare industry generates about 30% of the world's data volume, making it one of the largest contributors to global data generation, with an expected compound annual growth rate of 36% this year [1]. While humans cannot comprehend and work with such vast amounts of data, advances in the computational power of monitoring devices are producing datasets – and therefore insights such as atrial fibrillation detection - like never before [2].

The adoption of lifestyle fitness watches nearly doubled in the last five years [3]. These fitness trackers are capable of monitoring physiological parameters in all kinds of situations, is a clear indication of the increased focus on wellbeing among individuals. They are used to track vital sign parameters during resting, athletic activities and even sleeping [4]. However, most of these measurements do not meet institutional healthcare accuracy standards [5], but they are quite helpful in determining trends and overall health status. These are actionable insights that individuals can use to adjust their daily schedules, workload or sleeping habits.

Healthcare institutions have been using continuous monitoring systems for decades now, but wireless solutions are still not widely adopted [6]. Existing vital sign measuring devices are large, wired and cumbersome machines. Their high cost limits the number of patients who can be continuously monitored, typically restricted to the intensive care unit (ICU). Currently, the industry standard source for PPG signal is a fingertip sensor [7], measuring the common parameters like heart rate (HR), oxygen saturation (SpO_2) and heart rate variability (HRV). However, a wrist-based sensor is also a good alternative for measuring the same parameters and opens up many new feature possibilities, such as combining it with skin temperature. These solutions are not widespread yet in healthcare institutions but have the potential to monitor far more patients due to their relatively low cost.

This thesis investigates the specifications and considerations necessary for developing a wearable and wireless monitoring device capable of measuring oxygen saturation and skin temperature in a multi-patient system. The device needs to prioritize ease of use for the nurses, battery life relative to sampling frequency, and patient comfort, all while maintaining measurement accuracy.

1.2 Motivation

This research is based on a health-tech startup, Entremo's product, which is designed to help healthcare institutions and nursing staff to monitor multiple patients at once with a wearable and wireless setup, enabling them to detect deteriorating cases faster. I am keen on taking a more holistic approach in this investigation, exploring various aspects of the device development, as it is not only the technical specifications that make it viable on the market, but the entire package, including stakeholder considerations, design compromises and price points among many other criteria. As this product category is still novel in institutional settings, and there is no widespread adoption yet, the process of development becomes even more engaging and important. The advancement of such wearable devices have the potential to significantly improve efficiency in healthcare services, both in care quality and cost [8].

1.3 Purpose and relevance

Continuous monitoring systems enable hospitals to create very detailed and insightful patient profiles, thus increasing the quality of care they provide. Any changes in vital signs like oxygen saturation, skin temperature or heart rate are usually the first indication of a health status change and need close attention [9]. A recent study, which researched the diagnostic error rates for patients who needed to be transferred to ICU or died before transfer, concluded that a missed or delayed diagnosis occurred in 23% of cases, with 17% of these causing temporary or permanent harm. These errors were judged to have contributed to death in 6.6% of all deaths analyzed in the study [10]. Still, not including intensive care units, most European hospital wards do not currently have an automatic continuous monitoring system, this work is still carried out manually by nurses [11]. Manual measurements often take around 5 minutes per patient [12] and doing so for an entire ward means only a few data points per patient each day, thus limiting early detection capabilities.

In the spring of 2020, the outbreak of the coronavirus pandemic highlighted numerous inefficiencies stemming from this manual practice. The digitization of measurement data was conducted manually, knowledge-sharing between staff was irregular and sanitization protocols were often neglected, leading to very high infection rates among healthcare workers as well [13]. Another critical issue was brought to light as well, the shortage of nurses, which continued to worsen since the pandemic [14].

It was in response to these challenges that Entremo was founded, aiming to offer a comprehensive solution. Entremo's purpose is to assist the nursing staff in monitoring patients more quickly and effectively. By equipping patients with wireless wearable devices, nurses – rather than making measuring rounds – can dedicate more attentive time to patient care, access up to 100 times more data points[*] and receive alerts about any health deterioration, thereby reducing reaction times and late-detection complications.

There is economic incentive for the hospitals too, because slow transfer to ICU due to late detection of deterioration can lead to a 60% increase in costs [15].

Entremo's development efforts prioritize all three key stakeholders – patients, nurses, institutions – and aim to strike a balance in device design to meet all expectations and requirements of each group.

Beyond technical feasibility, the development of wearable medical devices raises important considerations of ethics, sustainability, and social impact. Continuous monitoring involves sensitive health data, which requires strict privacy protection and secure transmission. The use of batteries and plastics also introduces sustainability challenges, making durability, reusability, and energy efficiency crucial design goals. Finally, the broader social implications include reducing nurse workload and improving patient safety, while also ensuring equitable adoption and avoiding over-reliance on technology.

1.4 Objectives and scope

The objective of the thesis is to develop the product architecture of a wearable and wireless monitoring device that is capable of accurately measuring heart rate, oxygen saturation and skin temperature with wrist-based sensors. Although the industry standard PPG measurement is typically fingertip based [7], this work will introduce the decision factors behind selecting a wrist-based approach, which offers significant potential for new features and may eventually replace

[*] Manual measurements take place on average 3 times a day, whereas continuous measurement every 4 minutes totals 360 data points a day.

wired monitoring solutions. In addition to demonstrating the vital sign accuracy of the sensors compared to alternative devices, the research also encompasses other market viability factors like stakeholder criteria, multi-device system management, communication network choices and battery life expectations. Given the long and strict regulatory process of taking a medical device to market, it is not expected that the prototype will be used by actual patients in the near future. Therefore, the scope of this research is limited to in-house testing.

1.5 Goals

The goal of the thesis is to answer the question of what technical specifications and design considerations are necessary to build a wearable vital sign monitoring device that delivers meaningful efficiency and cost benefits over wired or manual patient monitoring practices. The work includes consideration of sensor placement strategies across the body (e.g. wrist vs. fingertip), compares measurement accuracy to established devices, and evaluates user experience to judge feasibility in non-intensive clinical setting.

Research questions:

1. To what extent can a wrist-worn device reliably measure heart rate, oxygen saturation, and skin temperature – under typical clinical conditions – compared with established monitoring methods?
2. What design and workflow choices enable smooth, low-effort integration into everyday nursing routines while maintaining patient safety and hygiene?

Reliability is defined by clinical grade error standard levels. Clinical conditions assume mobile, but predominantly slow-motion movements, room temperatures and indoor lighting conditions.

1.6 Research Methodology

This thesis follows a pragmatic and applied approach, combining quantitative and qualitative methods to address both technical feasibility and user acceptance. Quantitative validation against reference devices was used to assess measurement accuracy of heart rate, oxygen saturation, and skin temperature. At the same time, qualitative stakeholder interviews with nurses, doctors, and patients provided insights into usability, workflow integration, and patient comfort. A purely experimental laboratory study was not chosen, as it would neglect the social dimension of adoption, while a purely qualitative study would not provide the objective accuracy evidence required. The mixed-methods strategy therefore reflects the dual aim of this research: to establish whether wrist-based sensing can achieve clinically meaningful accuracy, and to identify the design choices that enable adoption in healthcare settings.

1.7 Delimitations

This thesis is limited to the development and proof-of-concept validation of a wrist-worn prototype measuring healthy volunteers. It does not include clinical trials in hospital wards, regulatory approval steps, or advanced calibration studies such as hypoxia testing for oxygen saturation. Other potential features, such as blood pressure estimation, respiratory rate, or fall detection, are recognized but remain outside the scope. The focus is therefore on demonstrating feasibility and identifying key design choices rather than delivering a clinically certified device.

1.8 Structure of the thesis

The thesis is divided into three main parts, a theoretical introduction that provides the necessary background knowledge, a comprehensive explanation of the procedures and technical solution, and finally, a presentation of results and conclusions.

In Chapter 2, the theoretical introduction serves as a foundation to provide sufficient background knowledge on vital sign monitoring devices in the healthcare industry, as well as the technical and physiological principles of PPG and temperature sensors. This section is necessary for the reader to understand current industry standards and to follow the thought process behind the chosen methodology and the specific details of the technical implementation.

Chapter 3 describes the entire process of the product architecture development, including details of the design considerations, technical hardware decisions and sensor data processing of the monitoring device.

In Chapter 4, the prototype's performance will be evaluated based on several metrics and the discussion of these results will address the research questions.

Chapter 5 presents a discussion, the project's conclusion, future outlook recommendations offering insight into the possibilities and limitations of the developed wearable device under market conditions.

2 Background

This section lays the foundation for the necessary background knowledge on vital sign monitoring devices in the healthcare industry, including the technical and physiological principles of PPG and temperature sensors. Background on the Entremo hardware development, battery characteristics and wireless communication are also presented.

2.1 Host company

Entremo is a health-tech startup founded by five Hungarian engineers with the aim of accelerating the adoption of digital solutions in healthcare environments such as hospitals and nursing homes. The need for continuous monitoring received unprecedented attention due to the coronavirus pandemic and many companies were formed to tackle the new challenges arising from the sudden changes [16]. Entremo was one of these companies, ready to work on new, affordable and more efficient processes of vital sign monitoring. Instead of wired devices, the focus was – from inception – on completely wireless and wearable solutions that are more suitable for a multi-patient fleet management system with an easy charging and assignment workflow. Lifestyle fitness trackers were already proving sensor capabilities [5], so Entremo's objective was to adapt and enhance these capabilities so that they can be used specifically in healthcare institutions. Such a new process in a hospital ward may increase the volume of patient data and insight, offer an alleviating solution to the nursing staff shortage, and improve institution economics as well [17].

2.2 System architecture

The Entremo system is a complete end-to-end remote monitoring device service that is designed to be operated by the nursing staff in a clinical environment, as seen in **Figure 2.1**. The most important vital signs to measure are heart rate, oxygen saturation, skin temperature and heart rate variability. In non-intensive departments, where there are no robust machines, the nurses equip the patients with Entremo devices, which continuously measure these physiological parameters. Multiple devices can operate simultaneously and independently within a ward, enabling staff to quickly view the status of all patients. The measurements are processed and transmitted to nearby communication hubs, which forward the encrypted data to the servers. The results are then displayed on Entremo's online dashboard, where nurses and doctors can view a live summary of all the patients and receive alerts about any deterioration based on pre-defined thresholds. As a result, changes in health status can be detected earlier, allowing for timely precautions and interventions. The device fleet is fully reusable, similar to other battery-operated equipment, making it highly efficient in terms of unit economics.

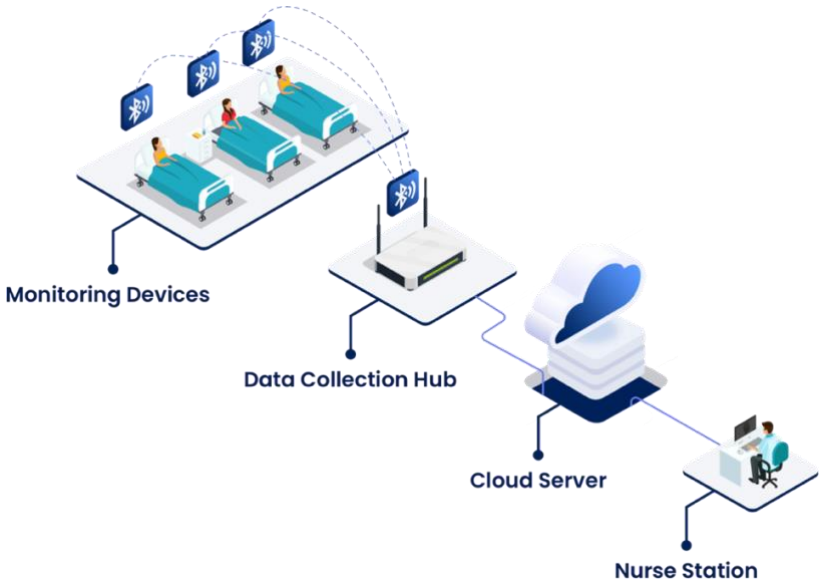


Figure 2.1: Isometric view of Entremo's system. Source: Entremo Business Plan

2.3 Medical monitoring systems and device types

The healthcare field is one of the leading industries in terms of safety management. The development of medical devices follows a much more rigorous and risk-averse direction compared to other fields.

In the European Union, the regulatory framework for medical devices has changed significantly in 2021 with the introduction of the Medical Device Regulation (MDR) 2017/745 [18]. It aims to improve transparency, device traceability and post-market surveillance – factors that contribute to maintaining public trust in the healthcare system.

Regulatory compliance means that every medical device in the EU must be audited and certified to prove adherence to strict safety and quality standards, ensuring human safety in operation. The complexity of a device directly impacts the time and cost required for regulatory approval. Moreover, changes made after an approval also need new audits or in some cases, complete re-certifications. It is no surprise that companies are always looking for ways to reduce their development costs by simplifying parts or building on existing solutions.

Vital sign measuring devices come in many forms and variations. The following section provides an overview of the various types currently available on the market.

2.3.1 Multi-parameter machine

The multi-parameter machine seen in Figure **2.2** is one of the most powerful tools in patient monitoring. Roughly the size of a shoebox, it is capable – as the name suggests - of measuring several types of vital signs, including blood pressure, electrocardiogram, temperature, oxygen saturation, heart rate, respiratory rate and more. The advantage of the machine is its high measurement accuracy, as the sensors are not bound to a single body part; each parameter can be measured at its optimal anatomical location. The detailed wave forms and measurements are displayed on its screen, so caregivers see live data right next to the patients. The machine is often mounted on a wheeled pole, allowing nurses to move it easily between patients. However, a limitation of the machine is that nurses need to manually operate it to take measurements.



Figure 2.2: Example of the multi-parameter machine and the rolling setup.

Additionally, its wired design restricts patient mobility, as the machine must remain connected during use. Given this device's expensive price, it is not economically viable for non-intensive care units to provide one for every patient [19]. As a result, this solution does not support continuous data capture.

2.3.2 Portable fingertip pulse oximeter, non-contact infrared thermometer (NCIT)

Healthcare institutions cannot place multi-parameter machines besides every patient, because it would be too expensive. Therefore, in most wards, the nurses are measuring the patients manually with several separate devices, as illustrated in **Figure 2.3**. They typically use a portable fingertip pulse oximeter to measure HR and SpO_2, followed by an infrared thermometer, a blood pressure cuff and they may even manually count breathing frequency if needed. This linear process takes on average 5 minutes per patient, which means that in a ward of 15 patients with 2 nurses, measuring everyone once takes at least 40 minutes [20]. Given the length of this cycle coupled with the nurses' additional duties, only about three rounds per day are realistically possible. Another limitation of this manual process is the need for data input. Because it is not an automated digital system, the nurses have to manually enter the measurements into their system or track them on paper in case there is no system, introducing further inefficiencies. The manual process also carries an increased risk of infection if proper sanitization protocols are not followed [20]. Devices are used sequentially on multiple patients, and although they are supposed to be sanitized between uses, this does not always occur. However, like the more robust multi-parameter machines, these individual devices also produce high-accuracy data points as they measure from dedicated, independent body locations.



Figure 2.3: Wireless pulse oximeter (left) and infrared thermometer (right). These devices offer great flexibility in mobility but lack automatic data integration into patient records.

2.3.3 Wearable devices

The third type of remote monitoring devices are wearables, which are worn by patients for a longer period of time. There are many different shapes and sizes on the market, based on the location of the data capture. The most widely used locations are fingers, wrists and the chest area [21].

These devices are designed to operate automatically, without any nurse intervention, so they require longer battery life and seamless connectivity to a data management system that stores and displays the results. Power is supplied either by disposable battery cells or a rechargeable battery pack. Either way, this is another responsibility for nursing staff to monitor battery levels. However, they spend less time with data management, because these devices usually transmit the data over the air through secure communication channels. These wireless monitoring setups thus yield up to two orders of magnitude more data points of the patients' statuses and enable many new use-cases for data analytics, which can all improve the quality of care. Being wireless, it is also much more convenient for the patients to be able to move around and leave their beds. Due to the discreet form factor, measurements can even be taken while patients are sleeping. Since these data management systems use some kind of alert system, this way nurses can receive alarms of deterioration even during the night, making detection much faster, possibly saving lives. The obvious disadvantage of wearables compared to robust systems is lower accuracy. With size and computing constraints, these devices have lower precision, but there are already leading examples [22] that demonstrate their ability to meet industry standards. As accuracy continues to improve due to advancements in processing capabilities, wearables are expected [23] to become increasingly common in clinical settings.

There are many companies worldwide, both big and small, who are developing vital sign monitoring devices for different use-cases. Some of these companies have similar development strategies as Entremo, with overlapping sets of parameters and intended use-cases, making them direct competitors. Being informed about their technological advancements and development direction is essential to Entremo's development strategy as well, so the following section will lay out the public information published about them, gathered from articles, patents and PR communications.

2.3.3.1 Healthcare wearables

Biobeat

Biobeat is an Israeli technology company developing two product lines in continuous patient monitoring [24]. As shown in **Figure 2.4**, one is a single-use chest patch designed for clinical settings, the other is a smartwatch-style wrist device intended for personal homecare use. Both devices measure a wide range of vital signs and offer a battery life of up to five days. They use an Early Warning Score system to detect patient deterioration. Both devices are available on the global market and have received Conformité Européene (CE) marking as well as approval from the U.S. Food and Drug Administration (FDA).

Oxitone

Oxitone is a company from Israel, developing their wrist device, Oxitone 1000M, which is designed for homecare monitoring to provide supportive supervision for primary care and follow-up processes [25]. As seen in **Figure 2.4**, they employ a unique sensor placement over the ulnar bone on the top of the wrist and hold patents and certifications for both the US and EU markets. Compared to other devices, the promised 24-hour battery life is relatively short. The device has a screen given that it is for personal home use, which could be a reason behind the short operating time. Another possible reason is that, instead of performing advanced processing on the device itself, it may transmit a large volume of raw data to their external servers. Recently the company has

begun developing a clinical reporting service system, as well as a personal data analytics and decision support platform.

Caretaker Medical

Caretaker Medical is a US medtech company founded in 2014 producing their VitalStream™ continuous patient monitoring device designed for many clinical use-cases from long-term acute care to perioperative care in the operating rooms [26]. They developed a wrist-based device that has a non-reusable finger probe attachment as well for better PPG signal accuracy. The device is about twice the size of other similar wrist devices but runs only for 12 hours. For mass-monitoring purposes, they have the VitalStation™ solution, a centralized data display system, which aggregates the data from their devices and helps the nursing staff in clinical efficiency. VitalStream™ is also FDA-cleared for cardiac output and non-intensive blood pressure monitoring.

Masimo

Masimo is a large medical technology company in the US with a 35-year history of developing products ranging from non-intensive patient monitoring, through hospital automation to lifestyle wearables [27]. Their Radius product line is designed for clinical institutional settings with multiple types of devices. Some models feature finger probe attachments on wrist-based devices, and they also offer a separately sold patch for skin temperature measurement. Their wrist-worn devices with optical sensors are designated for personal lifestyle use.

Philips

Philips Healthcare developed a patch device as well - the Biosensor BX100 [28] - although there has been little recent news about it. It measures heart rate, respiratory rate (RR), electrocardiogram (ECG), posture and activity level. It is a single-use device running on a small coin cell. The biosensor uses Bluetooth to communicate with nearby hubs. As one of the largest medical device manufacturers globally, Philips is primarily focused on complex systems, which may explain the relatively limited emphasis on wearable devices at present.

Summary

As above introduced solutions demonstrate, there are many different variations on the market for continuous vital sign monitoring, but no single solution has achieved widespread adoption. As a result, even seemingly small differences between products allow them to address distinct use-cases within clinical settings. At Entremo, the goal of my work is to develop a wrist-based device capable of measuring HR, SpO_2 and skin temperature in a multi-patient environment, in a fully standalone configuration (no attachments). While some of these requirements are addressed by direct competitors, none currently offer a complete solution covering all aspects.



Figure 2.4: Biobeat patch (left) and Oxitone bracelet wearable devices.

2.3.3.2 Lifestyle wearables

An adjacent field worth mentioning is the market of lifestyle wearable monitoring devices. There has been a huge upward trend in personal wearable popularity, largely thanks to companies like Apple, Google and Samsung [29]. These devices remind people daily that their health data could play a

crucial role in improving wellness and early intervention. Over the years, the set of measurable vital parameters has expanded, with newer models now capable of tracking skin temperature trends - and soon, blood pressure or blood sugar levels too. An important distinction from medical devices is that these lifestyle wearables are not certified for clinical use and cannot be relied upon for diagnostics.

Athletes are also beneficiaries of performance tracking with Garmin chest monitors or Whoop bands [30][31]. Data capture is valuable not only during athletic performance but also throughout the recovery phase, thus supporting the case for 24-hour usage. Some devices, such as the Oura ring, also serve as stylish accessories [32]. The Finnish company's device is one of the sleekest trackers on market right now.

2.3.4 Impact of wearables in healthcare

In the course of the last decade, wearable healthtech went from being a cool gadget to having life-saving potential both in healthcare and lifestyle environments [33]. With COVID-19 arrived an immense appetite to accelerate the development of remote continuous monitoring devices and companies were in the spotlight to prove themselves. One notable example is the Oura Ring, making headlines after the National Basketball Association (NBA) bought 2000 units to offer voluntary monitoring to the league's players to detect coronavirus symptoms as early as possible [34]. Another important milestone was the Apple Heart Study in 2019, which led to the introduction of atrial fibrillation (AF) detection in the Apple Watch [35], [36]. Accident detection in the Watch also helps save lives by notifying emergency services with the precise location of the crash. But wearables have also had an impact in a wide range of other applications, including fertility tracking, drug effect monitoring, sleep analysis, electrolyte tracking, heat-related illness prevention and more [37].

2.4 PPG methods

The photoplethysmography (PPG) signal is a versatile, non-invasive optical technique used to measure various vital parameters including heart rate, blood oxygen saturation, respiratory rate, and even blood pressure estimation [38]. Based on the principle of light absorption changes due to blood volume fluctuations, PPG signals can be obtained through various methods, each suited to different applications and environments. The most common approach is the transmission mode, where light is passed through a thin body part like a fingertip or earlobe, with the light-emitting diode (LED) and detector photodiode (PD) on opposite sides. This method provides high signal quality but is limited to specific body locations. In contrast, the reflectance mode illustrated in **Figure 2.5** positions both the light source and detector side by side on the skin surface, measuring the reflected light. This allows for greater flexibility in sensor placement and can be used on various body parts. Both transmission and reflectance modes are commonly used for wearable devices in sport, wellbeing and clinical settings. The optical measurements have various parameters such as body location, light intensity, wavelength or sensor arrangements.

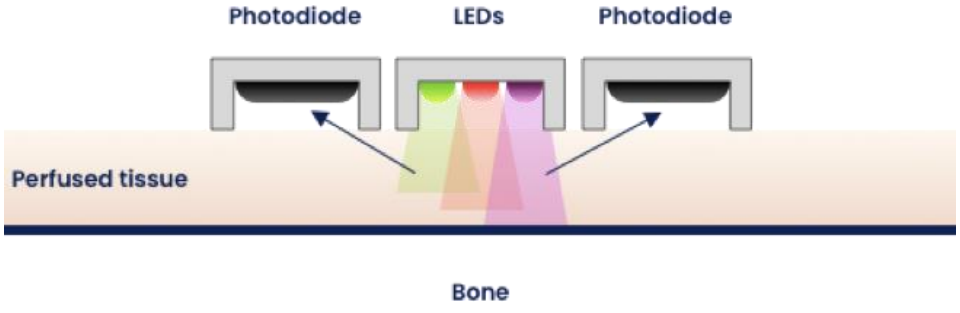


Figure 2.5: Reflectance-based PPG sensor principle, used by Entremo. Source: Entremo Business Plan

Another method to obtain the signal is called remote PPG (rPPG), which uses cameras to detect subtle skin color changes associated with blood flow, enabling non-contact monitoring [39], [40]. For example, using a smartphone's camera and flash to capture signals from a fingertip could offer widespread accessibility [41]. Beyond these, newer approaches include multi-wavelength PPG, which uses multiple light sources of different wavelengths to improve accuracy and provide additional physiological information. Wearable PPG sensors integrated into clothing or accessories offer continuous monitoring in daily life. Fiber optic PPG sensors provide electromagnetic interference immunity, making them suitable for use during magnetic resonance imaging (MRI) scans. Lastly, implantable PPG sensors, though invasive, offer the potential for long-term, highly accurate monitoring in specific medical scenarios, like research with animals [42].

The method of reflection-based PPG sensors begins with one or more LEDs emitting light, typically green (525 nm) red (660 nm) or infrared (940 nm), into the skin. This light penetrates the skin and underlying tissues to a depth of several millimeters, where it interacts with various components, including blood vessels, skin pigments, and other structures. As the heart beats, the volume of blood in the capillaries changes, altering the amount of light absorbed or scattered. Some of this light is then reflected back towards the surface of the skin, where a photodetector, placed adjacent to the LED(s), captures it. The result of the light modulation will be the PPG signal. The skin and tissue characteristics can significantly influence signal quality, as they vary depending on body location, thickness and pigmentation [43]. Even body hair can affect the optical path and reduce signal clarity. The process can be modeled with the Beer-Lambert law (Equation 2.1) [44], which is a relationship between the attenuation of light through a substance and the properties of that substance. It states that there is a linear relationship between the absorbance (A) and the concentration (c), absorption coefficient (ε) and optical path length (l) of a solution.

Equation 2.1:

$$A = \varepsilon c l$$

2.5 PPG artifacts

Another considerable challenge of optical PPG signals in wearable use-cases is signal noise caused by motion. Artifacts are distortions or disruptions in the signal that can significantly impact the accuracy of vital sign measurements. The most prevalent and challenging type is motion artifacts, which occur when there's movement between the PPG sensor and the skin, altering the optical path and blood flow dynamics. These artifacts can severely distort the PPG waveform, leading to inaccurate readings during physical activities or daily tasks involving arm and wrist movements. Other sources of artifacts include ambient light interference, poor sensor contact with the skin, physiological variations, and electrical noise from the sensor or surrounding electronics. Efforts to detect, remove, or compensate for artifacts - particularly motion artifacts - remain a critical area of research in PPG signal processing and wearable device development. Because waveform reconstruction from corrupted segments is rarely feasible, the most practical solution is often to discard the affected data and repeat the measurement after a brief interval.

2.6 Calculation of vital parameters

PPG signal can be used to calculate many medical parameters, thus removing the need to perform invasive methods for blood samples in many cases. For most applications the two most important parameters are heart rate and blood oxygen saturation, because they provide quite an indicative status level of the subjects, be that in a clinical, athletic or leisurely environment. Wearable devices make continuous measurements accessible to many, which leads to earlier detection of problems and more effective treatments [8].

2.7 Heart rate calculation

The heart rate can be calculated with multiple color illumination wavelengths, if the signal quality is good. The heart rate is the frequency of pulse waves in a temporal (time) domain. In a time-domain approach, heart rate is determined by identifying the main peaks of the PPG waveform using a peak-finding algorithm. The time difference (cycle length) between two consecutive peaks is measured, and the heart rate is then calculated by dividing 60 seconds by the cycle length, yielding a result in beats per minute (BPM). Alternatively, the signal can be transformed into the frequency domain, where the dominant peak within the 0.5-4 Hz range is identified. Multiplying this frequency by 60 provides the heart rate in BPM, within the physiological range of 30-240 BPM. **Figure 2.6** presents both methods.

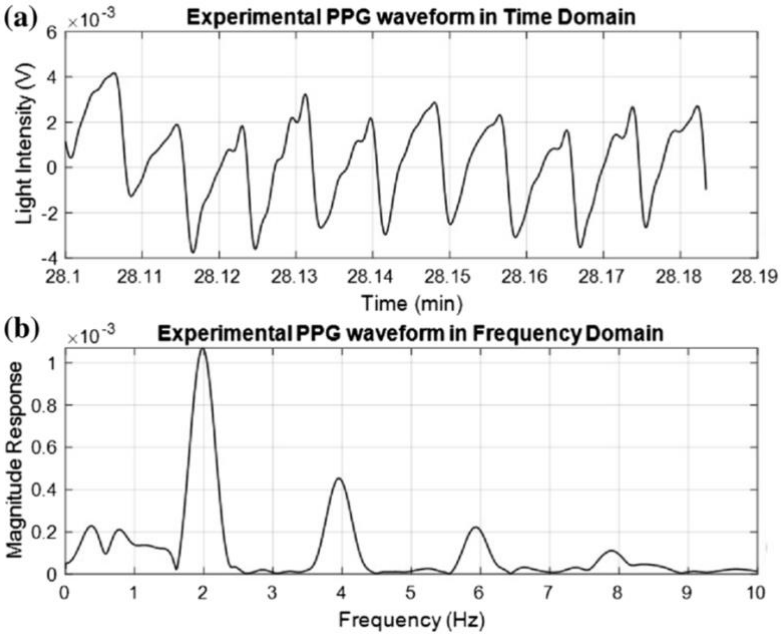


Figure 2.6: Peaks in time domain (a) and frequency domain (b) representations of PPG signal. Source: [45]

2.8 Oxygen saturation calculation

Oxygen saturation (SpO_2) is the ratio of oxygenated hemoglobin (HbO_2) to all hemoglobin concentration in the blood. For healthy people, this metric should be at least 95%, values below 90% are considered low, and readings below 80% may pose serious risks to organ function [46]. Calculating SpO_2 from a PPG signal employs pulse oximetry, which utilizes the different light absorption characteristics of oxygenated and deoxygenated hemoglobin, hence using the Beer-Lambert Law in the setup. The process begins with dual-wavelength measurements, typically using red (660 nm) and infrared (940 nm) LEDs [47].

The PPG signal is separated into a variable, pulsatile component of arterial blood (alternating current (AC)) and a static, non-pulsatile component (direct current (DC)) of the signal made up from the tissue, venous blood and non-pulsatile arterial blood, as seen in **Figure 2.8** The AC component is usually only 1-5% of the total signal, so after isolation from the DC component, it can be amplified [48]. Dividing the AC and DC levels of the two wavelength measurements, we get a ratio R (Equation 2.2). This R value is then mapped to an oxygen saturation level through a look-up table, which is the result of a desaturation study [49]. In these studies, oxygen levels in human subjects are artificially reduced from 100% down to approximately 75%, while being measured by

the test device. The results from the device are then paired with reference oxygen saturation levels to establish the R-to-SpO_2 mapping curve. A key limitation is that the resulting calibration curve can vary depending on the subject population, device configuration, and consistency of the test environment. Since desaturation studies do not go below 70% oxygenation, values in this range must be extrapolated - resulting in higher inaccuracy at low saturation levels. Most devices on the market apply a quadratic function for this mapping process, some examples are shown in **Figure 2.7**.

Equation 2.2:

$$R = \frac{AC_{red}/DC_{red}}{AC_{infrared}/DC_{infrared}}$$

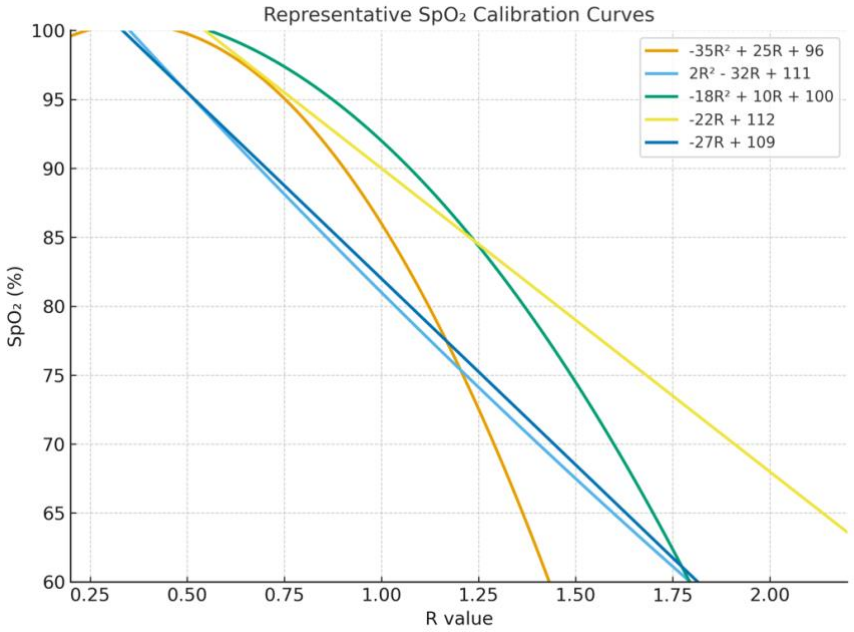


Figure 2.7: Example R curves for SpO2 calculation. The R curve calibration is a result of a desaturation study and varies depending on the characteristics of the study. In this thesis, a proprietary calibration study was out of scope and publicly available coefficients were used instead.

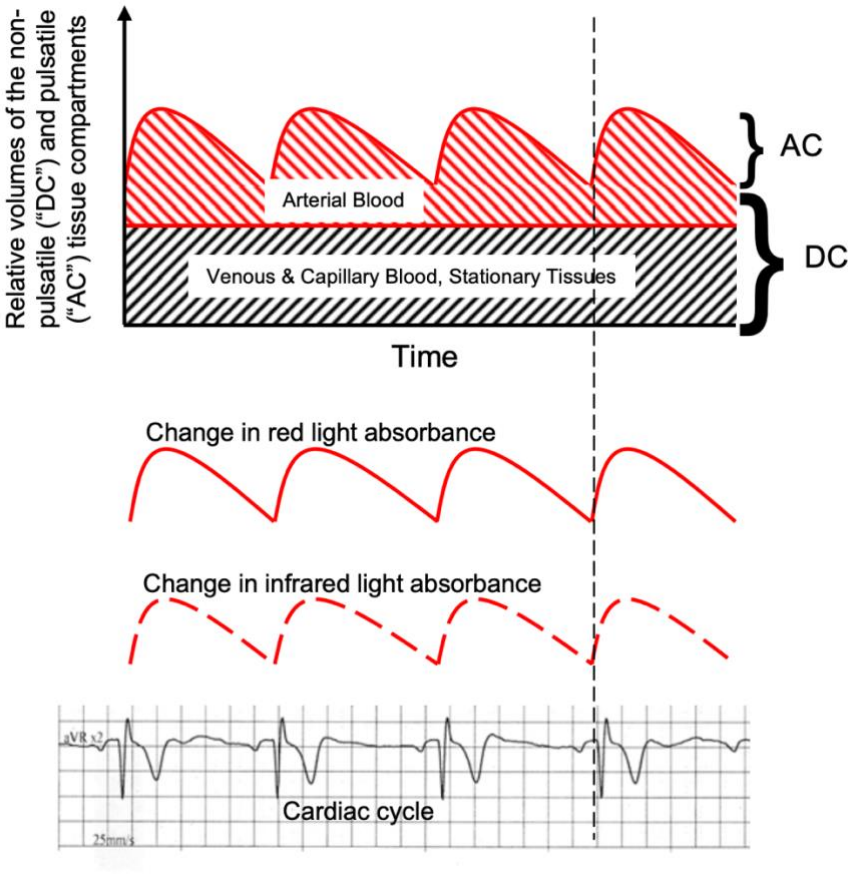


Figure 2.8: The amount of red and infrared (IR) light absorbed in the arterial compartment rises and falls with systole and diastole, due to the increase and decrease in blood volume. The volume that increases with systole is known as the pulsatile or AC component and the component in which the blood volume does not change with the cardiac cycle is known as the non-pulsatile or DC component. Figure and caption sourced from: [50]

2.9 Skin temperature

The skin, the largest organ of the body, - approximately 15% of total body weight – also plays as a vital role in thermoregulation, helping to maintain a body temperature of about 37°C [51]. It does so by regulating blood flow, sweating and radiation. Besides adapting to external forces like cold weather, a change in skin temperature can also mean the reaction to internal forces in the human body. A significant rise in temperature is most commonly the sign of the body fighting something, therefore body temperature is one of the most basic vital signs that is used for medical diagnosis and health management. The origin of today's thermometry dates back to 1868, when Carl Wunderlich published his study on temperature in the course of diseases, which included theories like the 37°C reference level, women's slightly higher temperature observation and diminution of average temperature with advancing age [52]. However, the use of body temperature as a diagnostic sign dates back to the earliest days of clinical medicine. Hippocrates (around 400 BC) recognized the importance of assessing body heat and taught physicians to detect abnormal temperature changes by touch. Precise measurement of body temperature was not possible until the development of thermal instruments in the sixteenth and seventeenth centuries in Italy [53].

There are several different ways to measure skin temperature, each with different advantages. One of the most basic methods is using a glass thermometer with mercury in it, which expands uniformly with heat changes, thus moving up and down on the visible scale. This method is rather slow and fragile, mainly used in immobile or external circumstances, so it's not ideal for medical

practices. Another method is the non-contact infrared thermometer gun, which is sensing thermal radiation emitted from the body. Ambient temperatures near the subject may cause measurement errors though, but it is fast and works for any object, so it is widely used in healthcare environments. Wearable devices use a third kind of method, which is contact thermometry with tiny (3mm) sensors, which rely on conductive heat exchanges between the skin and sensor. Since this technique measures the temperature of the sensor itself, it is crucial that it is in equilibrium with the skin temperature, else results will be inaccurate. It is therefore essential to maintain sufficient skin contact without modifying the immediate environment and that the sensor shall be shielded from the heat of the device and printed circuit board. Once these key variables and conditions are met, this method offers many advantages: being low-cost, small in size and able to do continuous measurements at various body parts, even under clothing [54].

It is important to note that skin temperature, which typically ranges between 33°C – 37°C, is lower than the core temperature ranging 36°C - 37.5°C [55]. The difference is more significant in limbs and extremities, so evaluation of the measurements will depend on which body part it was taken. Common measurement locations include core-proximate types like the inner ear and oral, and skin sites such as the temple, forehead, chest, armpit, wrist or finger. Wearable devices use the latter two locations, which usually produces lower temperatures, so without accurate evaluation algorithms, companies refrain from outputting exact measurements, they only display variability compared to the benchmark. In normal circumstances, this is already enough to detect patterns in sleep cycles, menstruation cycles or rapid physiological deteriorations [56].

2.10 Hardware layout

During the prototyping phase, a special development unit was created from scratch - through lots of research, experimentation and 3D printing - to make the testing and development more efficient. This development tool demonstrated in **Figure 2.9** contains the same electronics components as the proposed final device, but it doesn't have space constraints or battery limitations, as it's powered via a PC. As such, its design focuses solely on functionality and processing speed.

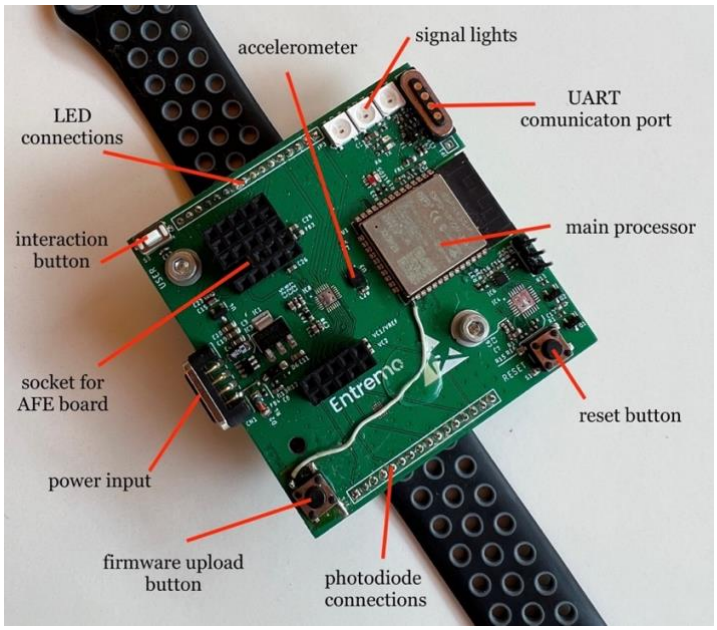


Figure 2.9: The PCB of the development device for signal and layout experiments, labeled with its main components.

The architecture set out requires a controller to handle the sensors, the power supply and the communication with a computer. This controller is the main processor on the development PCB, which controls the ADPD4101 analog front end (AFE), responsible for operating the LEDs and the samplings of the photodiodes; handles the serial communication interface and power supply through a magnetic USB; and operates the rest of the sensors and LED indicators. The AFE model was chosen for the benefits of 8 LED input connections, ambient light rejection and flexible sampling programming. As it can be observed in **Figure 2.10**, the PCB is mounted onto a 3D-printed watch base, allowing for experimentation with different sensor placements and measurement layouts to evaluate accuracy.

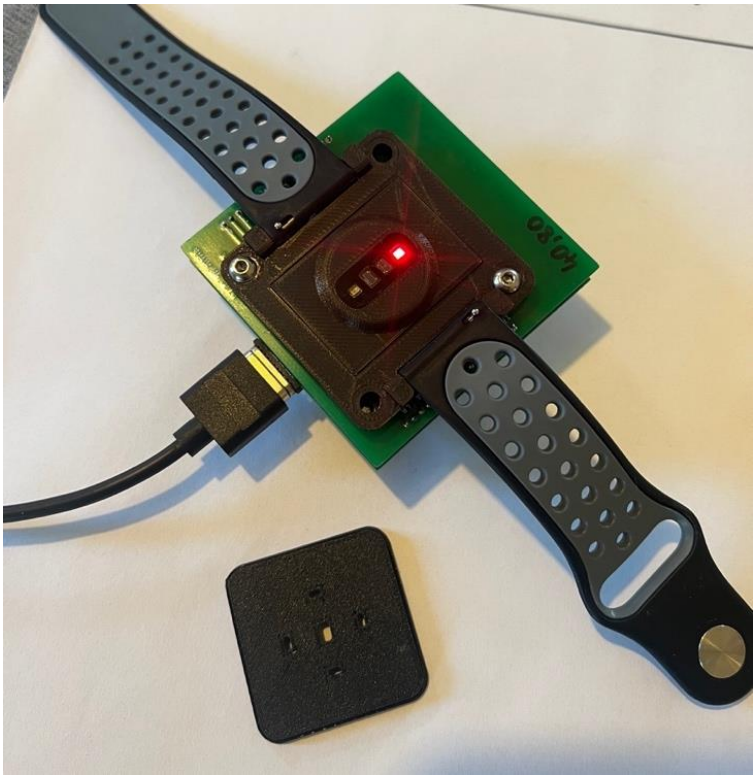


Figure 2.10: The bottom of the development device while calibrating, and below a sensor layout module that is interchangeable.

To find the most suitable LED sensor arrangement, the different layout holder plastics can be swapped and tested quite easily. As shown in **Figure 2.11** configurations varied in the number and positioning of LEDs and photodiodes, the distance between components, and even 3-dimensional characteristics. Some designs even included a spherical protrusion to explore the effects of improved skin contact.



Figure 2.11: More than 30 sensor layouts were tested on signal quality, robustness to noise and hardware cost. Layout 30 and 31 proved to be the most effective. The layouts were 3D printed, which enabled fast prototyping cycles.

2.11 Battery characteristics

Due to the wireless nature of wearable devices, they of course need some form of battery to power the electronics inside, including the microcontrollers, sensors, LEDs, displays, etc. The choice of battery must satisfy multiple constraints in design, size and power, so it requires a delicate balance between performance and practicality. Form factors of wearables are usually quite unique and are adapted to body parts and use-cases. Compactness and thinness are obvious design priorities as they directly impact user experience and ease of use. Finally, computing power and battery life are always expected – at times even demanded - to improve over time, which means that stagnation can significantly harm marketability.

There are three different types of batteries (**Figure 2.12**) that are usually considered for consumer devices:

- Multi-cell battery pack is a collection of cylindrical cells encased together and is used, when high charge capacity is needed, and the product has space to spare.
- Coin cells are useful for compact and low power consumption solutions. They are non-rechargeable, so without the need of a charging port, devices are easier to waterproof.
- Lithium-ion polymer (LiPo) batteries are the most popular for wearables due to their high energy density and design flexibility, which allows them to be shaped and sized to fit unique device geometries.



Figure 2.12: Comparison of the major types of batteries in wearable and consumer devices. Source: Fictiv

LiPos have the advantages of

- high energy density (100-140Wh/kg),
- small and variable sizes (270 Wh/L),
- high operating voltage (3.6V, 3.8V),
- wide application temperature range (-10°C-45°C),
- low self-discharge rate (6-10%/month)
- long cycle life (300-2000 times)[57].

As a result, LiPo offers a long battery life and fits well into thin architectures like smartphones or watches. However, LiPo batteries also pose a higher flammability risk due to the softer 'pouch' casing and flammable electrolytes inside. Therefore, additional safety measures must be implemented, including deep discharge protection, load regulation, short-circuit protection and water resistance[58]. LiPo batteries (**Figure 2.13**) have their own Protection Circuit Module (PCM) board and thermistor as well to provide another layer of protection. It is also advisable to allocate extra space within the device enclosure to accommodate swelling that may occur as the battery ages.

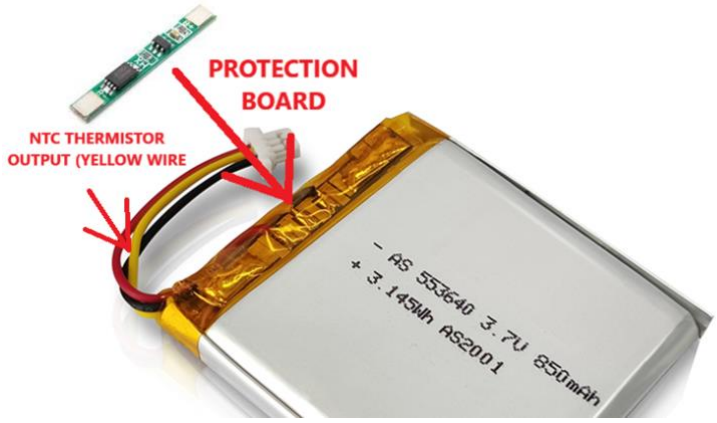


Figure 2.13: Example of a LiPo battery with its PCM board and a negative temperature coefficient (NTC) thermistor, which can read the battery temperature. Source [58]

Next-generation batteries are already being explored to reach higher energy density and lower flammability. Lithium-oxygen (or lithium-air) batteries have a higher density than current LiPo versions, but still have many challenges to overcome such as cycle length or instability[59].

2.12 Wireless communication

The intended purposes of the Entremo system require a fully wireless communication solution, which includes communication between the device, hub and server as well.

2.12.1 Communication between device and hub

For the devices, a low-power communication technology is needed, such as Bluetooth Low Energy (BLE) or LoRa.

BLE is short-range, typically only up to 100 meters with Bluetooth 5.0 and walls considerably reduce effective range [60]. With its ultra-low power consumption, it is ideal for applications that have frequent updates. Operating on the unlicensed 2.4 GHz ISM (industrial, scientific, medical) band, BLE has data rates up to 2 Mbps, which is beneficial for continuous data transfer and high-bandwidth communication. Scalability of the network is limited because of its peer-to-peer nature, but BLE modules are relatively cheap.

LoRa, as its name suggests, is a long-range solution that can cover distances up to 5 kilometers in urban settings [60]. Its extremely low power consumption is suitable for infrequent long-range transmissions, resulting is potentially years of battery life. LoRa uses the 868 MHz ISM band in Europe and has extremely low bandwidth (0.3 – 50 kbps), making it unsuitable for high-throughput or real-time monitoring scenarios.

Based on this comparison, BLE emerges as the optimal communication interface for wearable devices in a local network. In the case of Entremo, devices communicate through BLE with proprietary receiver hubs (**Figure 2.14**) that are supposed to be setup in patients' rooms, corridors and nursing stations. The devices send the data packets to the closest hubs, therefore patients can be mobile even during measurements. Even if no hubs are close by (e.g. the patient goes for a walk in the garden), measurements are saved to the internal memory and are sent once connection is available again.

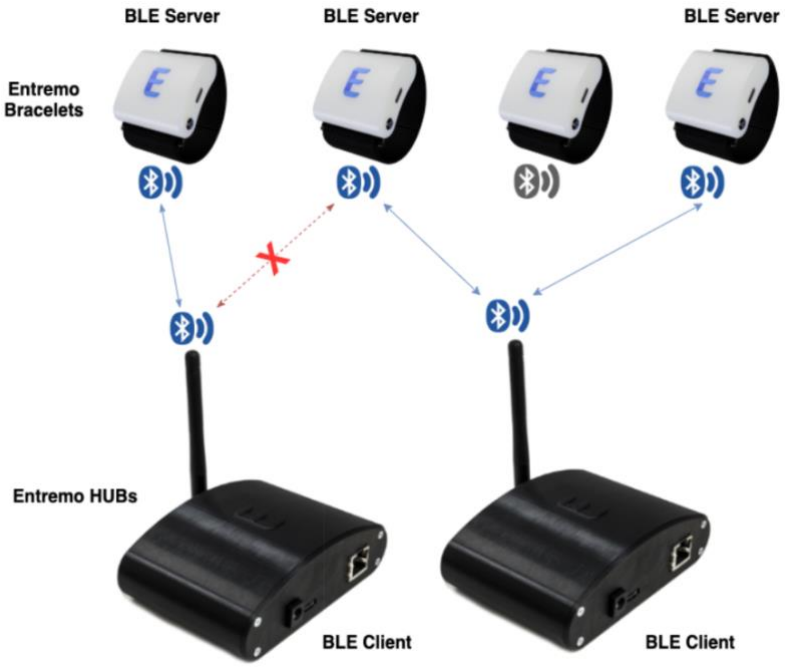


Figure 2.14: A BLE client can connect to multiple servers, but a server can only serve one client. In the Entremo ecosystem, the bracelets are the servers and the hubs are the clients. When unconnected bracelets

have data to send, they are producing advertisement packets which are broadcasted on one of three different radio channel frequencies. Hubs are constantly scanning for available bracelets, and immediately connect to the next available bracelet to receive data packets.

The protection of personal information is a top priority. Accordingly, the wearable devices and hubs do not store any patient-identifying information; they only collect and transmit physiological measurement data. Patient assignment to measurement data occurs exclusively on the backend server. Still, given the sensitive nature of the transmitted data, encryption and authentication are needed. Currently, data packets are encrypted with a 128-bit Advanced Encryption Standard (AES) algorithm, for which the secret key is factory programmed into the devices and hubs. This approach simplifies implementation, but the system can be hacked if a device is jailbroken or the key is leaked. Device authentication is based on the manufacturer data field, the BLE MAC address and shared secret key.

2.12.2 Reducing data volume

In the case of a wearable device development, power consumption aspects always need to be prioritized. Radio interface – BLE communication in our case – is one of the most energy hungry components of the device, where needs are proportional to the amount of data sent or received. Improving power efficiency therefore involves minimizing data volume and configuring the radio to operate with minimal overhead.

Data volume can vary depending on institutional use-cases. It is up to the medical staff to determine which parameters to measure and at what frequency. Evidently, the fewer parameters and smaller frequency, the more energy a device can save. Additionally, if the staff don't require raw data of measurements like full PPG waveforms, then further energy can be conserved by doing signal processing and outputting results on the device itself. While, this method requires more compute power in the device, it significantly decreases the amount of data sent wirelessly.

Another way to downsize is to use a compression algorithm. The two main types of compression techniques are lossy and lossless. Lossy compression reduces data size by permanently eliminating some parts of it, which are deemed less important or redundant. This method is commonly used with multimedia formats. For sensory data, it would be hard to determine what part of it could be eliminated and missing vital sign measurements could prove worrisome. In contrast, lossless compression retains all original data, enabling full reconstruction during decompression. This is essential when data integrity is critical – though the trade-off is lower compression ratios compared to lossy techniques. The two techniques can be combined to have a hybrid solution as well, as presented in an ECG study [61], however it leads to more complex development and audit processes.

2.12.3 Communication between hub and server

The hub has a SIM card module and connects to the internet via 5G through a safe virtual private network (VPN) tunnel. If cellular connection or a SIM card is not available, the hub also has an Ethernet port that can be used. The hub forwards the measurement data in Concise Binary Object Representation (CBOR) format to a UDP/CoAP server, which translates it to JSON format and pushes it to the backend server. This server is responsible for managing all the data collected and entered by users through the dashboard. It is implemented with Django and the Django Rest Framework (DRF) with a Postgres 12 database, consisting of 6 tables, presented in **Figure 2.15**. In the current version, everything is stored in the same database, which is not optimal as the index can grow unnecessarily large, slowing down queries. In the long term, measurement data will have to be moved to a separate time-series database, such as TimescaleDB.

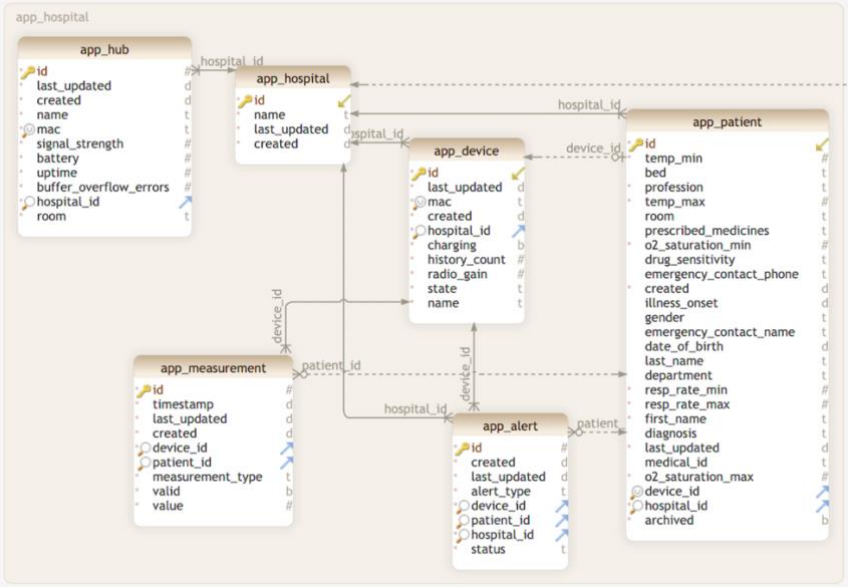


Figure 2.15: The Entremo backend database structure with its 6 tables for the different objects: hub, device, hospital, patient, measurement and alert.

The staff in the healthcare institutions will be able to see measurements and manage patient data through the Entremo dashboard application seen in **Figure 2.16**. After login authentication, they can manage the patient list by adding new patient profiles, see the statuses of devices and hubs and receive alerts of the deteriorating cases, which moved beyond the pre-set thresholds.

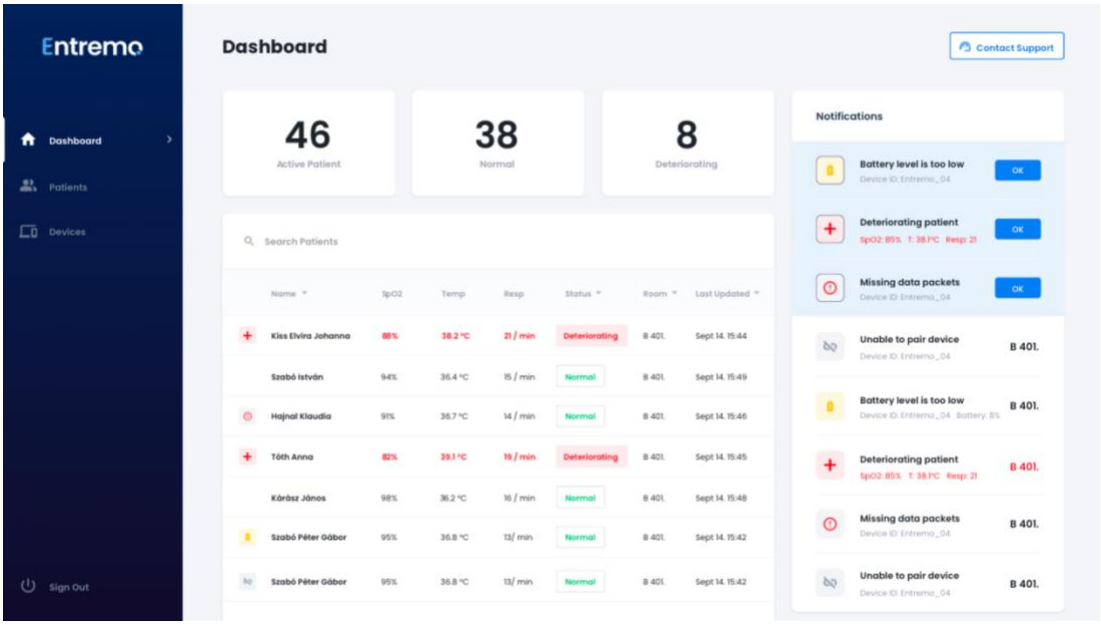


Figure 2.16: The Entremo dashboard, where staff is able to monitor and manage patient data.

3 Methodology

This section describes the methods implemented in this research project based on the research plan. The guiding principles and requirements for the design process are presented. Data processing steps and techniques are also outlined.

3.1 Research plan

The goal of this project is to establish what specifications are required to build a wireless vital sign monitoring prototype that is tailored to clinical setting and achieves industry standard accuracy* in HR, SpO_2 and skin temperature measurements.

First, I conducted hour-long online and in-person interviews with nurses, doctors and past patients to turn their needs into measurable targets for accuracy, comfort, hygiene, and upkeep of such a monitoring device. Those targets anchored the main design decisions regarding form factor, materials used, charging system and LED/photodiode settings. Then the development environment was setup to produce repeatable and standardized signal quality tests to tweak the sensor settings. With the groundwork in place, I compared fingertip and wrist measurements under the same protocol to see how they differed in amplitude, motion sensitivity, and usability. Once ideal sensor arrangements and calibration were defined, I tuned the high-quality data capture signals in a signal-processing pipeline: I filtered out drift and high-frequency noise, computed signal-quality indices (DC level, peak-to-peak amplitude, kurtosis, motion), and kept only segments that passed SQI thresholds. On that foundation, I calculated the vital signs: heart rate from peak detection, SpO_2 from the red/IR AC–DC ratio with R-to-SpO_2 mapping, and skin temperature from a calibrated sensor with offset calibration. Finally, I conducted benchmark tests in real world situations in parallel with established reference devices and created a comparison analysis to determine accuracy metrics.

3.2 Stakeholder interviews

In order to develop something that the market actually needs and can use, I found it important to assess the needs of all kinds of stakeholders that will come into contact with the Entremo device. This approach helps reduce the risk of allocating resources to unnecessary or unwanted features. For easier presentation of the preferences of the stakeholders, their answers will be categorized as either hardware or software related.

3.2.1 Preferences of nurses

I have conducted interviews with several nurses in Budapest to hear from the end-users themselves what is important to them. Nurses are the primary operators of these devices, so it is quite necessary to provide a smooth and satisfactory process for them, which is able to help their workflows significantly. As for physical attributes regarding the hardware, their first thought was that the device should not be an obstacle in any of the current tasks they are carrying out with the patients, like taking their blood pressure, eating or going to a restroom. They therefore prefer a wireless solution, which does not bind the patients to their beds and there are no extra cords in the way. Nurses also expressed concern about patients tampering with the devices and accidentally changing settings or operations. Hence the decision not to have any buttons on the bracelet. Patients need zero interaction with it and cannot change anything by mistake. A recurring question from the nurses was about battery life. They would prefer at least a couple days of uninterrupted operation

* Industry standard error acceptance is MAE < 3 BPM for HR, RMSE ≤ 3% for SpO_2 and MAE < 0.1°C for temperature.

and also voiced their wish to have a very convenient charging system, which only makes their lives easier and not harder. Meaning, that they want to see at first glance if a device is sufficiently charged or depleted and do not want to operate dozens of plugs in their nursing room. Another important factor they brought up was sanitization. In healthcare institutions especially, all reusable equipment needs to be sanitizable so that they are safe to use for the next patient. There are several ways of sanitization with varying levels, from wiping with an alcoholic tissue to sustained submerging in disinfectant liquid. In a busy hospital ward, where several patients would use the same device in a day, it is quite important to have a fast and convenient sanitization method. The potential for mismatching devices with patient profiles was another concern. Nurses stressed the importance of a streamlined and fail-safe pairing process. Based on this feedback, the pairing mechanism was tied in with the charging system. After the pairing is done, nurses also want personalizable settings for each patient, because measurement frequency and parameter thresholds can vary based on individual conditions and status. Based on these custom thresholds, nurses want to be notified as early as possible if someone is showing a declining status. Several nurses shared anecdotes about late detections, where due to extreme shortages of staff, some of these complications were only discovered hours later, sometimes too late to save a patient's life. Therefore, an alert system should consider the likely movements and whereabouts of the nurses and have multiple ways of drawing attention, possibly through visible and audible channels as well. Some nurses mentioned another frustration: when measuring the vital signs with separate devices not connected to their systems, measurement results need to be written down on paper and later entered into the digital patient record system. It is no surprise that they would welcome any solution that could do this automatically, without their involvement.

3.2.2 Preferences of doctors

Doctors have the least amount of contact with such a monitoring device, therefore in their answers they focused more on systematic aspects rather than details of the operation. It is their responsibility (ultimately the medical director's) to decide which hospital wards should use such a system, they can be quite different in characteristics after all. According to doctors, the four most relevant wards in this case are cardiology, internal medicine, post-operation and the emergency care unit. These wards have patients, whose conditions can change rapidly, and vital signs are usually the first indicators of such a change. In most cases, doctors will not be near the patients when these deteriorations happen, so it is no surprise that they prefer a system, where they have the option to view the measurements remotely too if they want to. The viability of this is also dependent on the patient record system's security and server setup by the institution.

Such a solution enables the use-case of home monitoring too. Some doctors said that they would see a benefit in giving certain discharged patients a monitoring device to use for a two-week follow-on period.

Another recurring preference was the ability to measure blood pressure (BP). There are already studies and devices set out to measure BP in a cuffless way, but this would not be part of Entremo's offering.

3.2.3 Preferences of patients

Finally, patients were surveyed too to find out what their preferences and experiences were in hospital wards. The number one factor among them is comfort. In case they would have to wear such a device for maybe even multiple days, but at least multiple hours a day, they of course prefer something that does not cause them irritation. It is expected that patients will remove the bracelet in certain situations, such as showering, so the device must include an automatic detection feature to determine whether it is being worn. This helps prevent false or misleading measurements from

being recorded during those times. On the other hand, if the device remains worn during activities like handwashing or showering, it must be water-resistant. Proper sealing and material selection are necessary to ensure the device shell can withstand splashes and brief exposure to water.

Almost all respondents said that they are familiar with the concept of fitness watches, so it doesn't surprise them that vital signs can be monitored with them, or that they need to be charged periodically. It is beneficial to have hospital equipment that resembles familiar consumer products, because it reduces patient anxiety, making the hospital environment feel less intimidating.

Another input from patients who have slept with some kind of monitoring device before, is the annoyance of continuous light flickering. They meant not the red and green light of the PPG sensors, but the outward facing status light, which signaled active measuring. Since the Entremo device is designed without a screen or physical buttons, indicator lights are the only source of feedback. As a result, their behavior must be carefully designed to strike a balance between reassurance and comfort. The indicator should give peace of mind to both nurses and patients that the device is functioning, but in a subtle way so that it doesn't bother the patient all day and night.

Lastly, several older patients - especially those more mobile within their rooms - expressed concern about falling without being noticed or being unable to call for help. Multiple anecdotal stories were shared in which elderly individuals spent hours on the floor before receiving assistance. As a result, a fall detection feature that can notify staff immediately would be highly welcomed by this demographic. This is definitely on the future feature list of Entremo, but the current version of the device will not yet support it.

3.3 Design considerations for the device hardware

Based on the interviews I conducted with the different users and stakeholders in the monitoring process, I was able to formulate a product strategy for the Entremo device. In this section, I will outline the hardware-related design considerations that guided the development of the monitoring device.

What will the device do?

The purpose of the Entremo system is to automate the vital sign monitoring process in healthcare institutions, so that nurses and doctors have orders of magnitude more data points about the patients' health statuses, which enables higher precision care. The monitoring device continuously measures HR, SpO_2, RR and skin temperature, uploading the data to a cloud-based system accessible to medical staff even remotely.

Who will be wearing the device?

Based on feedback from doctors, the most suitable hospital wards for Entremo deployment would be cardiology, internal medicine and post-operative care. Most of the time, these environments have patients in stable condition, often mobile to some degree. They are usually staying for multiple days if not weeks until they are discharged. Therefore, the device should be comfortable, light and fairly unnoticeable for constant wearing and even when sleeping. However, measurement accuracy cannot be sacrificed for comfort, so the method of securing the device in a fixed position is necessary. The selected plastic strap meets this requirement; its rugged material securely holds the device in place while allowing for comfortable, long-term wear.

How is the device operated?

The device is designed to be handled solely by the nursing staff, in order to avoid any user errors or disruptions in the measurements. For this purpose, there are no buttons on the device, the measuring process starts and stops automatically based on sensor data that checks if there is skin

contact or not. There is also no display screen on the device for a number of reasons. First, a screen that continuously shows measurement results may cause unnecessary concern for patients, undermining their sense of calm during their stay. The goal is to minimize patient anxiety by limiting their interaction with or awareness of the device. Second, removing the screen significantly extends battery life, as displays are among the most power-consuming components. Third, omitting a screen also reduces manufacturing cost, making the device more economically viable for widespread deployment. As a result, without buttons or a screen, the nurse must assign the device to a patient profile via the central dashboard interface. From that point on, all processes are automated and seamless, requiring no further input from the patient.

What are the status light signals?

Even though everything is automatic, there still needs to be some kind of feedback from the device about its current status. Therefore, there is an LED light on top of it, which signals whether it is charging, fully charged, in pairing mode, measuring mode or in an error state.

Should the device be wireless or wired?

The majority of monitoring devices that have some kind of wired part do have the advantage of not needing a battery or wireless connection, and these can measure with higher accuracy if the PPG sensor is on the finger and wired to the wrist device. However, Entremo aims to develop a more seamless monitoring process, where the patient is not bound to their beds and there are no wires dangling around them. By having a fully wireless device, we enable patients to move freely within their rooms or in the institution. Mundane activities like eating and sleeping are also easier without any wires in the way.

Should the device be reusable or single-use?

The single-use scenario is only feasible with a cheap enough component like a patch. This solution includes a standard processing unit with the sensors, which can be mounted into a patch. The Philips Biosensor BX100 is such a device, designed to be placed on the chest area. This method would limit the range of measurements Entremo set out to include, so it was fairly easy to commit to the reusable device avenue. The most important requirement this decision brings is sanitization. Between different patients using the same device, disinfecting must be quick and reliable.

What materials to use?

Several factors must be considered to ensure the biocompatibility of the device so that adverse side effects can be avoided during the use of it. Material selection needs to satisfy a wide range of people with different ages, skin compositions, hair levels and arm sizes. The shell of the device is made of vapor smoothed Polyamide 12 material, which is gentle on the skin. Additionally, it is also lightweight, ensuring comfort even over extended use. The plastic strap is flexible and specifically designed for skin contact, as it's a standard watch strap. Both the shell and strap are waterproof and easy to sanitize with an alcoholic wet wipe.

Charging port

For the sake of simplicity, the device only has a single port, which is for charging. When choosing a connection interface, the requirements were to have an intuitive connection flow, USB charging and data transfer functionalities and the option of being waterproof. A magnetic connector seemed to be a good idea for its easy use, so the choice of connector became the HytePro M512 (**Figure 3.1**), with 4 pogo pins, which requires no precise alignment and resists unintended disconnections.



Figure 3.1: HytePro M512 magnetic connector and cable [62]

Charging system

The Entremo system is designed for parallel multi-patient monitoring, so the charging workflow needs to fit into this model as well. In order to reduce the nurses' tasks with charging cables and storing the devices, a case seen in **Figure 3.2** was designed that can charge 5 devices at the same time, without needing to touch any cables. The 5-slot charging dock has fixed magnetic contacts and is powered by a single 5V input. The devices drop into the magnetic bay and begin charging automatically. Disconnecting it is quite easy by tilting the device from its place, triggering pairing mode automatically, enabling nurses to assign the device to the intended patient profile.



Figure 3.2: The Entremo charging station, capable of holding 5 devices at the same time.

3.4 Developmental data capture setup

In order to test the capability of the proposed sensors and develop the algorithms to output vital sign parameters, the development board (**Figure 2.10**) was used on human subjects to provide the raw signal recordings. It was a low-risk setup, without any invasive procedures, with careful attention to safety. The plastic strap was only tightened as much as it was still comfortable for the subject, no electronic parts contacted their skin, no heat-up was likely because of low LED energy

settings, and since the framework was powered by a PC through a USB port, no lithium battery was attached. All participants were fully informed about the conditions of the test setup, including any potential risks. Participation was entirely voluntary and based on informed consent. No personally identifiable information was collected - only raw PPG and temperature signals and the date of recording. These signals cannot be directly or indirectly linked to any individual. No prior medical records or other personal data were gathered.

For consistency, several guidelines were followed during all measurements to provide the same conditions:

- The subject is sitting in a chair, with their arm resting horizontally on a flat table surface beside them, palm facing down.
- The measuring device is put on the top side of the wrist, 1-2cm away from the ulnar bone bump. Firm skin contact is necessary, so the exact strap hole used and wrist circumference is noted.
- During the measurement, the subject is calm and stationary, without sudden moves. In case of movement, thus causing motion artifact, the recording needs to be discarded.
- Measuring in 15 second intervals, ten times overall. Between each recording, the device is taken off and put back on to minimize the impact of the placement location.
- Room temperature should be around 21°C, so that no sweating takes place, lighting conditions should be uniform throughout the recordings.

In this setup the device is connected to a PC to save the recordings, which limits the scenarios this framework could be used in. So, for additional testing of robustness, another version was created to be tested in real-life scenarios. This version includes an SD card to enable wireless measuring in any kind of setup for multiple hours. Once the device is returned, the data can be downloaded and processed. With this modification I was able to create a reference setup, where additionally to the Entremo prototype, a Polar H10 chest strap monitor was also used to monitor multiple hours of vital sign activity. This chest strap only measured HR with electrical signals and provided only the output results, not the raw data, but the two parallel recordings could be aligned with the timestamps, and comparing accuracy was made possible.

3.5 Clean data requirement

In this section, a brief overview is given of the data capture parameter settings and expectations that determined whether a signal capture should be used or discarded.

- *Sampling frequency:* Depending on the use case a 25 to 500 Hz sampling rate is suitable for PPG measurement, where the lower end is suitable for HR measurement only, and the higher end is for PPG waveform shape analysis. In Entremo's case, we measure HR and SpO_2, but need to be conscious about data volume and battery operation, so a sample rate of 125 Hz is ideal. This is fixed for all measurements. The difference between each sample rate level is shown in **Figure 3.3**.

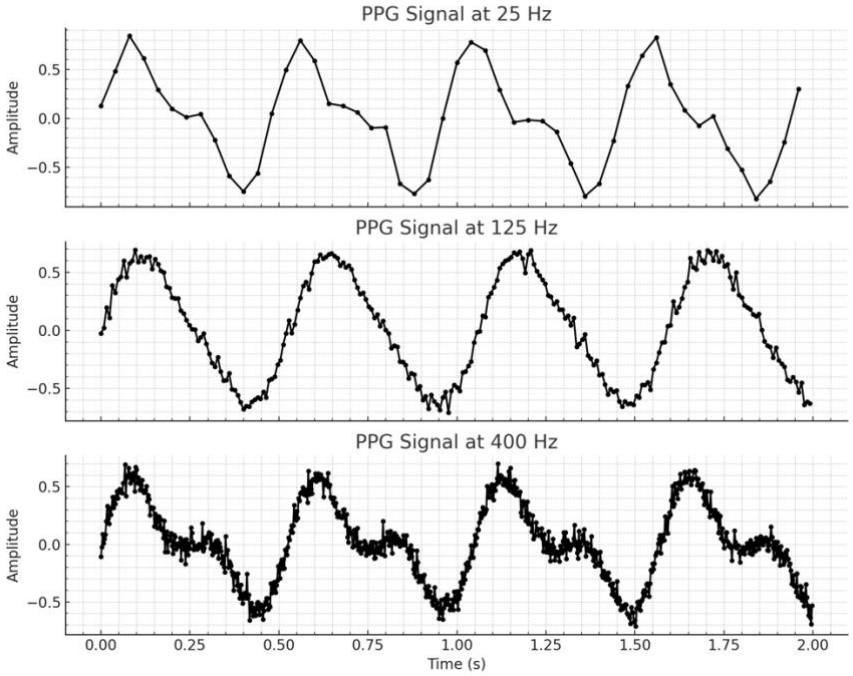


Figure 3.3: Difference in signal resolution at 25, 125 and 400 Hz.

- *Calibration:* Before the sensors start measuring, a calibration step is run, where the LED power and amplifier gain settings are set, which depend on e.g. ambient lighting, signal noise or skin characteristics.

- *Discard conditions:* Only those signal captures are kept and processed, which are considered clean recordings. So, in case of significant motion artifact spikes, constant signal oversaturation, too small wave amplitude, baseline wander or signs of physical obstructions the recording is manually discarded and repeated.

- *Signal output:* the output from the PPG signal recording will be 3 channels: green, red and infrared. For HR calculation, any one of them can be used, while SpO_2 is calculated from red and infrared together.

- *HR range:* Results should be in the range of 30–180 BPM as the subject is stationary and relaxed.

- *SpO_2 range:* Only healthy subjects were measured so results should be in the range of 90-100%, anything below that is likely an error in recording.

- *Temperature range:* Results should be in the range of 35-37°C.

3.6 Comparing wrist and finger measurement locations

In the product design of the Entremo device, a decision had to be made whether to measure PPG signals on a finger or on the upper wrist area. The decision would affect several aspects of the device, such as form factor, accuracy metrics, comfort, hardware complexity or practicality.

The finger location provides a significantly better PPG signal with higher amplitudes and less motion artifact, because the finger has a richer blood perfusion and less affected by muscles or tissue thickness as opposed to the wrist location. If only PPG measurement is the goal, then the finger-based approach is the optimal; it is no surprise that healthcare institutions use fingertip pulse oximeters as shown in **Figure 2.3**. However, when other parameters are also needed to be measured and wireless design is a priority, all other comparison metrics tip the scale towards a wrist-based approach. There are devices like the Caretaker VitalStream, which connect a finger sensor to the wrist device with a wire, but this makes it more cumbersome for nurses to handle the devices and also increases the complexity of the device. At Entremo, the decision was made to favor simplicity in form factor and user experience, while striving for sufficient medical grade sensor output.

Nonetheless, experimental efforts during development included measurement with a 3D printed finger-based prototype, shown in **Figure 3.4**, for several positive benefits. When testing sensors or layout arrangements, it proved helpful to reduce the number of uncertainties in the process as much as possible and use the best measurable signal possible. So before conducting tests on the wrist, the finger measurement was used to create a clean dataset for algorithm experimentation and baseline for benchmarking. These preliminary tests helped with the later validation of wrist-based measurements to determine what the cause of a poor performance could be. For example, if the sensor design worked well on the finger, wrist issues may stem from mechanical fit or contact pressure.



Figure 3.4: Iterations of the 3D printed finger-based prototypes used for early development and benchmarking. The same sensors were used in it as the wrist version.

3.7 Signal processing

Signal processing plays a crucial role in extracting meaningful data from the raw measurements collected by wearable sensors. Since these sensors often produce noisy signals due to factors such as motion artifacts, sensor imperfections, and environmental influences, effective filtering techniques are essential to improve data quality. Filters are employed to reduce high-frequency noise, remove outliers, and smooth data, ensuring that the signals reflect the true physiological state. The following section discusses the signal processing techniques used, their rationale, and the impact they have on the performance of the Entremo device.

3.7.1 Filters

Filters are mathematical tools to modify or manipulate a signal in order to allow or suppress certain frequencies. In general, there are two main types: low-pass and high-pass filters. The combination of both types results in band-pass and band-stop filters. One of the key parameters in filter design is the cutoff frequency, at which the filter begins to significantly attenuate the signal (transition from allowing frequencies to blocking them). The steepness of this transition is the order of the filter and as it gets higher, more complexity and latency are introduced.

Filter design requires delicate balance between performance and efficiency, so it is always tailored to the specific needs of the application. In this project, the MATLAB filter design tool was used to calculate the parameters of the filters applied to the PPG and temperature data.

Low-pass filter: allows frequencies below the cutoff frequency to pass through while attenuating higher frequencies. It is useful for removing high-frequency noise such as muscle artifacts or electronic interference.

High-pass filter: works in the opposite way of a low-pass by allowing frequencies above the cutoff and attenuating lower frequencies. It serves to eliminate DC bias and baseline wander caused by movement or slight sensor contact changes during the measurement.

Band-pass filter: combines a low-pass and high-pass filter to allow only a specific frequency band to pass. It was used to isolate the HR's frequency range of 0.5Hz – 4Hz.

Median filter: it is a non-linear method used to remove impulsive noise and outliers as illustrated in **Figure 3.5**. It is effective for removing spikes and salt-and-pepper noise while preserving the edges of the PPG signal. The algorithm takes a series of data points with a given window size, calculates the median for the series and replaces the value in the center of the window with the median, as defined in Equation 3.1. The window is typically chosen to be odd, so there is always a central point in it. The size of the window determines the performance power of the filter. In our case, a window size of 5 was used to remove motion artifacts or spikes due to sensor errors. This is the recommended size for the 125Hz sample rate we used to have balance between noise removal and preserving the underlying HR signal.

Equation 3.1:

$$y[n] = median\left(x\left[n - \left\lfloor\frac{windowsize}{2}\right\rfloor\right], \ldots, x\left[n + \left\lfloor\frac{windowsize}{2}\right\rfloor\right]\right)$$

Where, y[n] is the filtered signal and x[n] is the original signal.

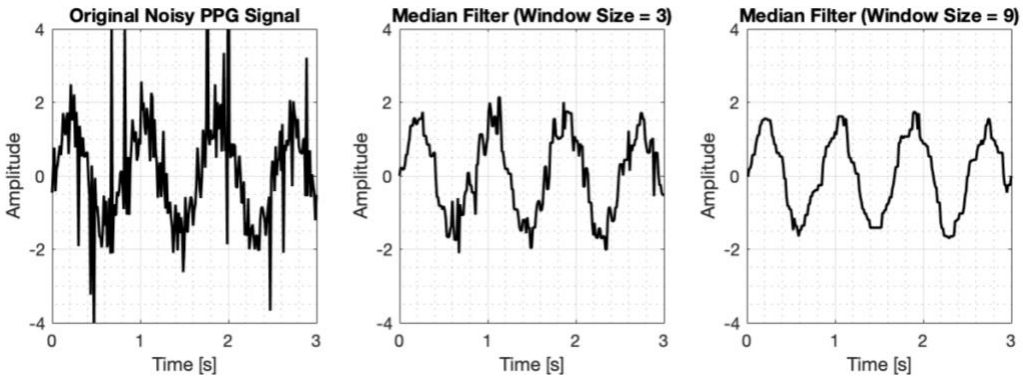


Figure 3.5: Effect of applying a median filter with different window sizes.

Moving average filter: it is one of the simplest and most commonly used types of linear filters in signal processing. Operating on a sliding window of values, the filter computes the average within the window. Its function is to smooth the signal by removing high-frequency random noise. It is very easy to implement and is computationally inexpensive. The filter can blur the sharp edges, hence it is not recommended for signal with rapid changes. The moving average filter, defined in Equation 3.2, behaves as a low-pass filter, where the larger the window-size, the lower the cutoff frequency will be. A window-size of 5 sample proved sufficient in this case as well.

Equation 3.2:

$$y(n) = \frac{1}{windowsize} \sum_{d=0}^{windowsize-1} x(n-d)$$

3.7.2 Signal quality metrics

When analyzing PPG signals, several signal quality metrics can be used to determine the accuracy, reliability and effectiveness of the dataset for further processing. The metrics aid in the assessment whether the signal is sufficiently clean enough or it requires additional preprocessing.

The following indicators were derived from each signal recording to assess quality:

- **AC amplitude** (A_{AC}): AC is the pulsatile components of the signal are where heartbeat waves are, ranging from 0.5 to 4 Hz. After applying a bandpass filter, the mean amplitude of the waves is calculated.

- **DC offset** (A_{DC}): the non-pulsatile component of blood tissues causes a shift of a constant value, the DC offset. After filtering the raw signal with a 0.2Hz low-pass filter, the mean of the signal is calculated as the DC offset.

- **perfusion index** (P): it is the ratio of AAC to ADC to assess peripheral blood flow and pulse strength.

- **signal-to-noise ratio** (SNR): measures the strength of the signal relative to the noise.
- **kurtosis** (K): it is a statistical indicator of the sharpness or flatness of the peaks compared to a normal distribution (peakedness), so it is efficient in reflecting the presence of outliers. It is regarded as good discriminator between low and high-quality signals. MATLAB's built-in function was used to calculate kurtosis, defined by Equation 3.3, where x is the signal, μ is the mean of the signal and σ is the standard deviation.

Equation 3.3:

$$K = E\left[\left(\frac{x-\mu}{\sigma}\right)^4\right]$$

- **power spectral density** (PSD): shows how the signal is distributed across different frequencies, making it easier to identify if the signal contains low-frequency or high-frequency noise.

In addition to these metrics, another subjective indicator was used for signal quality assessment, a Poincaré plot. The diagram is essentially a scatter plot of successive points of the signal to visualize the variation. The common method in HR analysis is to fit an ellipse to the scatter of points. The more the ellipse is fitting, the cleaner the signal, as seen in **Figure 3.6**. As studied in [63], a Poincaré plot can help in the diagnosis of several different health conditions: determining classification of respiratory diseases, assessing heart-rate variability or classification of patient severity based on body temperature.

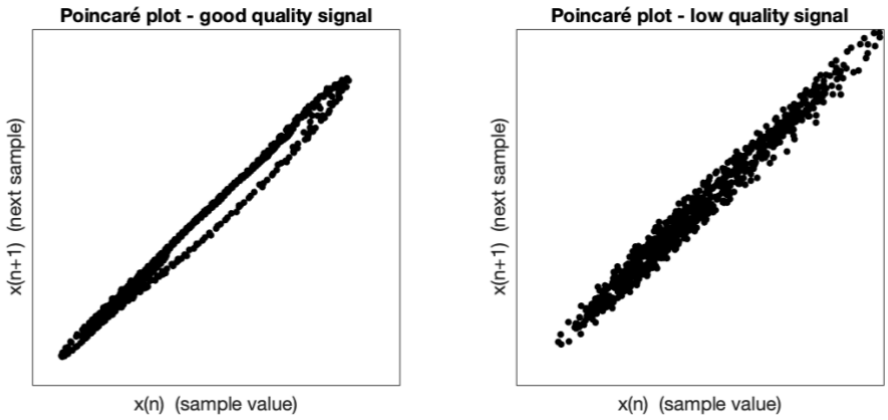


Figure 3.6: Comparison of ellipse fitting in Poincaré plots between a good and bad quality signal after being filtered. The opening in the center of the ellipse and cleanness of the left signal makes it easy to determine its quality from a glance.

The entire pipeline of the signal processing is presented in **Figure 3.7** below:

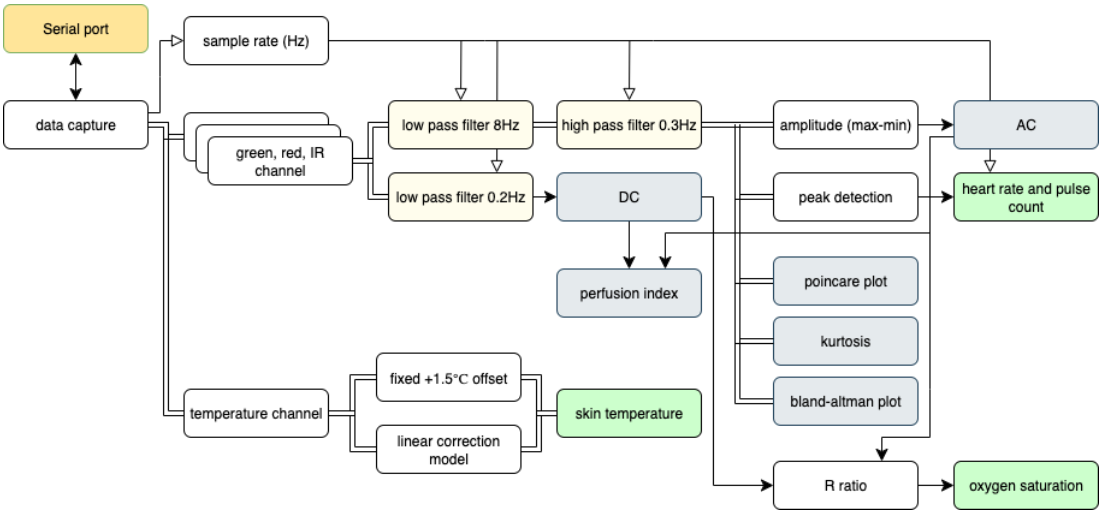


Figure 3.7: Block diagram of the processing pipeline for the Entremo prototype.

3.8 Heart rate calculation

Having preprocessed the signal with filtering and seeing satisfactory signal quality indicators, the data series was then ready to be analyzed for physiological parameters. Heart rate is calculated by finding the peaks of the heartbeat waves and averaging the intervals between them. For peak detection, the publicly available *peakdet* function was used, made by E. Billauer [64] This function

sweeps through the signal and finds a local maximum every time the signal moves at least *delta* above a previous local minimum and finds a local minimum every time the signal decreases at least by *delta* from the previous local maximum. **Figure 3.8** shows the top and bottom peaks found with this method. This essentially means that there needs to be a considerable amount of amplitude change from a previous peak to find another one. If this *delta* is too small, then every ripple will be categorized as a peak, and if it is too large, genuine beats won't be detected. There are several methods as to how to determine the size of *delta*; the most common heuristic was used in this work, choosing 20% of the amplitude of the signal. After detecting the top and bottom peaks, averaging the number of them results in the pulse count. Dividing the length (in samples) of the signal by the pulse count gives the average distance between heartbeats (d). The final HR value is calculated as $BPM = \frac{SR}{d} \times 60$, where SR is the sample rate of the signal and d is the distance in samples between the peaks.

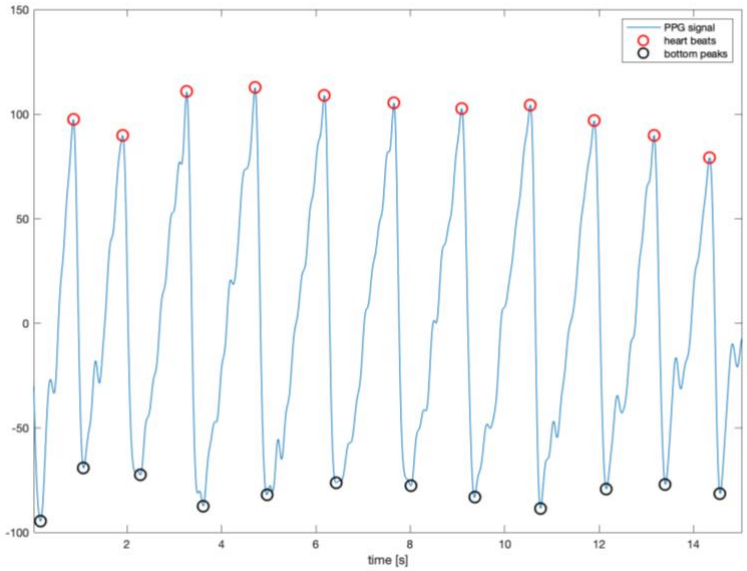


Figure 3.8: Top and bottom peaks found in a 15s PPG recording. Peakdet function used from [64] with *delta* =0.2 x $\text{amplitude}_{\text{minmax}}$; bandpass filter of 0.5 - 4 Hz, 5 sample median filter, 5 sample moving average filter, sampling frequency of 125 Hz.

The advantages of the algorithm used are its swiftness and simplicity, if the signal is clean enough, which makes it suitable for real-time analysis as well. Moreover, finding both maximums and minimums is also desirable as those will be used to calculate oxygen saturation levels and signal quality metrics.

In order to determine the accuracy of the Entremo sensor architecture, reference measurements were carried out during the entire development process with devices available on the market. Longer measurement tests took place during sleeping as well, where the test subject wore both an Entremo device and a Polar H10 chest strap the entire night. The results of the comparison are presented in Chapter 4.

3.9 Oxygen saturation calculation

Oxygen saturation (SpO_2) is the percentage of all hemoglobin binding sites in the blood that are occupied by oxygen and serves as a core vital sign metric to provide early warning in changing health conditions. For healthy adults the value is between 95 – 100%. A reading in the 90 – 94% range is classified as mild hypoxemia and often prompts supplemental oxygen or closer observation,

especially in patients with cardiopulmonary disease. Values between 80% and 90% signal moderate hypoxemia; at this level tissues are at genuine risk of inadequate oxygen delivery. Finally, an SpO_2 below 80% constitutes severe hypoxemia, a critical threshold associated with imminent organ dysfunction that mandates immediate action.

While HR calculation needs only one light source out of the green, red and infrared wavelengths, SpO_2 calculation needs both red and infrared sources. The *R* ratio of the pulsatile (AC) and non-pulsatile (DC) components, which absorb the lights differently, is defined in Equation 3.4 and is used for mapping to saturation levels.

Equation 3.4:

$$R = \frac{AC_{red}/DC_{red}}{AC_{infrared}/DC_{infrared}}$$

The values used for AC and DC were the previously mentioned A_{AC} and A_{DC} signal quality metrics. This *R* ratio is then used in Equation 3.5 along with c_n calibration constants, called R-coefficients. They are obtained once, in a controlled hypoxia study, by least-squares fitting *R* to a reference arterial oxygen saturation, and then hard-coded in every oximeter of that design. In the development of the Entremo device, such a hypoxia study has not been conducted yet, so the widely cited Nelson Wagner absorption coefficients [65] were used. This limitation naturally affected the accuracy of the device. The performance analysis compared to a reference oximeter device is presented in Chapter 4.

Equation 3.5:

$$SpO_2 = c_n R^n + c_{n-1} R^{n-1} + \cdots + c_1 R + c_0$$

3.10 Skin temperature measurement

Body temperature has been used as a health status indicator for thousands of years [53]. It signals infection and deterioration through fever and it is less influenced by emotional factors than other vital signs. Core temperature for normal levels ranges between 36 – 37.5°C and skin temperature is around 2–4°C lower, which also reacts faster to ambient changes [66].

The hardware layout of the temperature sensors was an available resource at the company, so despite not being a part of this research work, a brief description of it helps in the understanding of the process to achieve accurate readings.

In the Entremo device, four different thermometers were used: one to measure ambient temperature, one for battery temperature and two (an analog and digital) to measure skin temperature. The ambient temperature serves as a baseline reference and is used for skin contact detection too. If the skin temperature sensors drop within a threshold of 2°C from the ambient level, then most likely there is no contact between the sensors and the skin. Alternatively, PPG signal quality indicators such as A_{AC} flattening or A_{DC} drift can also be used to determine any changes in skin contact. In case skin contact is no longer present, incoming data points are flagged to display inactivity of the device in the dashboard system used by hospital staff. Battery temperature is a built-in component that is used for safety and charging regulation, it is not used in the processing of the skin contact sensors. The last two sensors were an analog LMT70 and a digital TMP 117 thermometer from Texas Instruments. In order to isolate the sensors from the PCB heat, a hole was cut around them and an opening from the bottom side into the sensor. Thermistor beads of the sensors were pressed into a silicone-based thermal paste to ensure minimal heat loss in the transfer.

In the other side of the paste are the high thermal conductivity aluminum pieces shaped as half-moons for each sensor, which are gently pressed into the skin for stable contact. An additional temperature sensor was also placed in the communication hub to measure room temperature, to allow for a heat-balance model accounting for external changes in the room, such as open windows, sunshine heat, etc.

Even with consistent and clean data capture, one of the key challenges of skin temperature measurement is how to map it to core temperature. There are several methods for such a mapping, but complexity is often too big of a compromise for a wearable bracelet:

- **fixed or adaptive offset:** applies correction based on empirical data and trend models, it is fast and requires few additional parameters, but inter-subject error can be larger and ambient changes can affect it
- **heat-flux:** uses a thermal pad to heat the skin and measures the heat flux, resulting in a highly accurate reading, but it is slow and requires bulky and power-consuming extra hardware
- **multi-node heat-balance modelling:** uses several physiological parameters to build a model, including subject-specific metrics that are not easily attainable; it is robust to activity and environmental effects, but draws more power than simple offsets
- **data-driven machine learning (ML) model:** trains regression on large dataset of pairs of measurements and reference core temperatures; good inter-subject adaptation but needs significantly higher compute

Considering the hardware and computing constraints of the Entremo device, only the correction offset method was a viable option at this point of the development. Since the trendline of the measurements closely followed the core temperatures, a linear correction offset was used that needed to be tested on different datasets too. Further analysis of the accuracy compared to a reference device is shown in Chapter 4.

4 Results

The following section presents the results of the product design and evaluates the performance of the data processing pipeline used to calculate the vital signs by analyzing the accuracy compared to reference devices.

4.1 Completed Entremo prototype

The purpose of this thesis was to present two significant parts of the process of developing a wearable healthcare monitoring device: user experience design and vital sign data processing.

4.1.1 User experience design

In a healthcare institution, there are multiple stakeholder categories to consider when introducing any kind of device into the system. The following needs were established and incorporated into the design based on feedback from the three stakeholder groups asked, namely nurses, doctors and patients:

Decisions based on nurse-driven needs as primary operators:

- Wireless and unobtrusive device - must not interfere with BP cuffs, meals, restroom visits, offer freedom of movement.
- No buttons/screens - eliminates patient tampering and necessitates seamless processes.
- Multi-day battery + easy charging – more than 48-hour runtime, quick and easy docking with charge-state visibility.
- Fast wipe-down sanitization - reusable shell/strap to withstand alcohol wipes.
- Fail-safe pairing and per-patient thresholds - pairing tied to charger, per- patient profile customization for alerts.
- Early, multi-channel alarms - visual + audible cues.
- Skin contact detection – auto-pause when device is off-wrist.

Decisions based on doctor-driven needs as clinical supervisors:

- Remote access to follow live trends - secure cloud dashboard with alert system.
- In-patient target wards - cardiology, internal medicine, post-operation, nursing homes.
- Cuff-less BP (future) - acknowledged but out of scope for this current research.

Decisions based on patient-driven needs as wearers:

- Comfort for multi-day wear - light strap, fixed but gentle fit.
- Water resistance - shower/hand-wash tolerance.
- Low-glare status LED - avoid sleep disruption.
- Familiar “fitness-watch” look - reduces anxiety and increases willingness to wear.
- Fall detection (future) - highly desired by mobile elderly people, but not in the scope of this current research



Figure 4.1: The final version of the Entremo device

The fully assembled product, incorporating these guiding factors, is shown in Figure **4.1**. The strap is an easily sanitizable comfortable plastic, which can be tightly secured for most wrist sizes. The only output interface is a magnetic Hytepro connector, which handles charging and optional data transfer too. On the bottom of the device, the sensors are placed in a protruding bump to maintain better skin contact. The LEDs and photodiodes are positioned in the center of the bump with a glass covering, flanked by two half-moon shaped thermometers. On the top of the device, the E sign is an LED status indicator, which signals charging, pairing, measuring and error states. The entire Entremo ecosystem is illustrated in Error! Reference source not found..



Figure 4.2: The Entremo ecosystem with the monitoring device, charging station, communication hub and dashboard.

4.2 Comparison with reference devices

After the development phase of refining device specifications, establishing SQI metrics for data capture and filtering signal, a real-life accuracy test was setup with commercially available reference devices for each vital sign parameter Entremo measures. The following sections will introduce the results of these tests by vital sign category.

4.2.1 Heart rate accuracy

During development, most reference measurements were carried out in controlled and calm conditions for no longer than one-hour sessions, because the development kit was not portable and wireless. This meant that accuracy of the signal data was already evaluated but lacking an extensive and real-world testing environment. When the hardware reached the phase of a wireless and compact formfactor, such real-world scenarios could be tested for performance. Both an Entremo device and a certified Polar H10 chest strap were placed on the subject for multiple nights of sleep. Results throughout the nights proved to be consistent. **Figure 4.3** displays the comparison of the two signals from one of the nights.

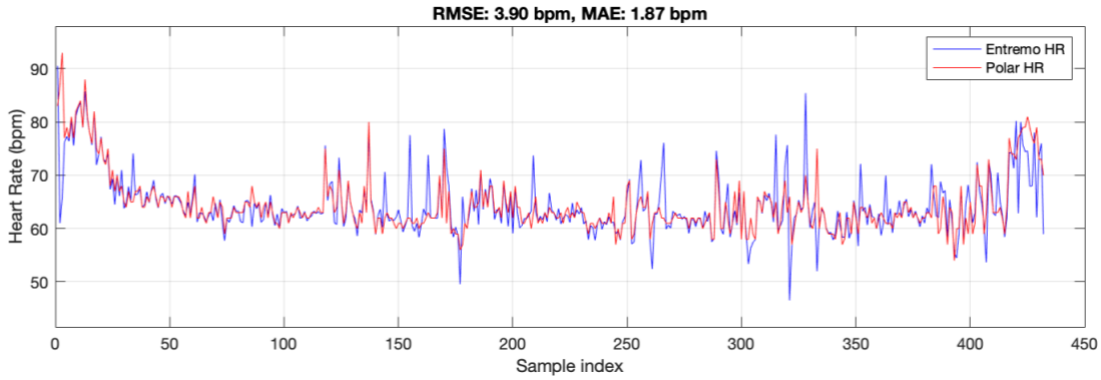


Figure 4.3: Continuous measurement from an Entremo and Polar H10 device with the subject sleeping. Sampling frequency: 125 Hz, HR datapoint frequency: 1 Hz, duration: 7.5 hours.

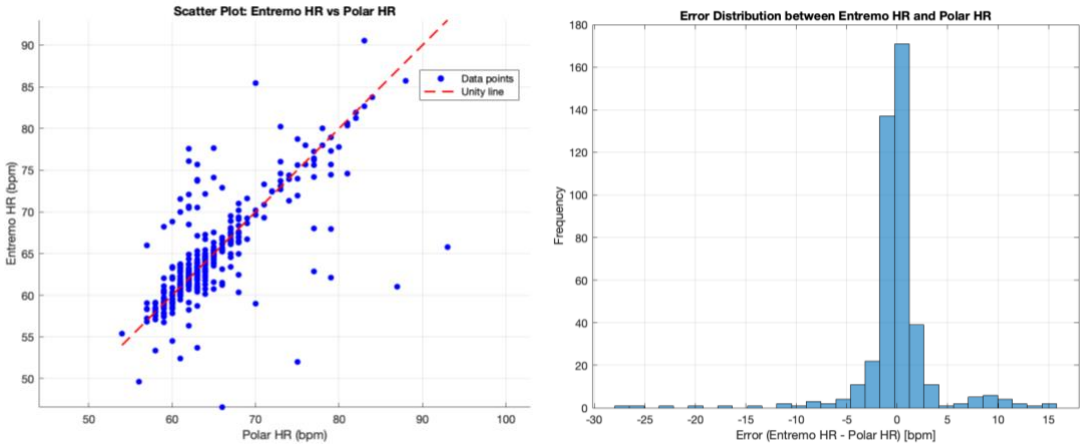


Figure 4.4: Scatter plot (left) of the comparison is displaying a strong linear agreement between the two devices. An error histogram (right) shows that bias is essentially zero.

It can be concluded that the Entremo device tracked the reference device very closely with a mean absolute error (MAE) of 1.87 bpm. This result meets the ± 3 bpm target for promising clinical use. Shown in **Figure 4.4** the scatter plot widening beyond 80 bpm indicates a higher variance with Entremo overshooting the reference points. This suggests that filter and peak detection parameters need further tuning for faster HRs. The root mean squared error (RMSE), which correlates to the size of the errors was 3.9 bpm. The error histogram in **Figure 4.4** centered at 0 bpm also clearly shows that bias is negligible and the algorithm does not systematically over- or underestimate heart rate. Bland–Altman analysis, which measures agreement levels between devices - showed that Entremo had a bias of -0.09 bpm, standard deviation (SD) of 3.9 bpm and 95% limits of agreement (LoA) of -7.74 to 7.55 bpm in relation to the Polar H10. The two devices agree without systematic offset present, but the range is large enough to warrant only trend level monitoring.

Due to the scope of this research, there were some limitations to HR calculation. Without a significantly large set of test subjects, measurements are not representative to an institutional patient population. While the feasibility and performance of HR calculation were demonstrated, a large-scale clinical validation is necessary in the future. Another factor unaccounted in this setup was the continuous movement of the test subjects. Even though sleeping contains occasional shifts in body position it is still far from walking or any other rapid movement activity that could influence the performance of the sensors. These motion artifacts and other signal noises could be mitigated by implementing sensor fusion models that include data from accelerometers, thermometers or different LED channels (e.g.IR channel instead of green).

4.2.2 Oxygen saturation accuracy

The accuracy test of SpO_2 measurements included seven - over 65-year old – test subjects over a two-week period, where they were asked to measure their oxygen saturation twice per day (in the morning and afternoon) with both the Entremo prototype and an industry standard finger-tip pulse oximeter. The analysis of the 150 paired measurements showed that the reference oximeter reported a mean saturation of 95.2% ± 2.2, while Entremo measured 95.9% ± 2.1. Bland–Altman analysis showed a bias of +0.71 percentage points, with a standard deviation of 2.91%, giving 95% limits of agreement ranging from –5% to +6.42%. The average absolute error was 2.31%, while the RMSE was 2.99%. **Figure 4.5** illustrates that most points fall within the agreement limits, with no clear proportional trend across the saturation range.

These results indicate that Entremo's performance is close to the clinical acceptability threshold: according to ISO 80601-2-61, pulse oximeters must demonstrate an RMSE ≤ 3% in the 70–100% range, which is narrowly satisfied in this dataset. While feasibility of SpO_2 monitoring with the Entremo prototype was demonstrated, due to the limitations of the project, it cannot be stated that measurement accuracy is satisfactory as of yet and it still needs refining for clinical use. In the scope of this test, only commercial pulse oximeters were available as reference devices, not the industry standard arterial blood gas analysis, which provides the most accurate results of saturation. The small sample size and dataset are also not representative of a hospital ward, no mapping model can be devised from such a small test. Finally, in order to achieve clinical accuracy for the full range of 70-100% of saturation, a hypoxia study needs to be carried out. In this study, healthy subjects based on skin-tone distribution undergo a controlled gas-induced saturation reduction to 70-80%, while test and reference devices measure the change. This results in representative mapping in the full range of saturation.

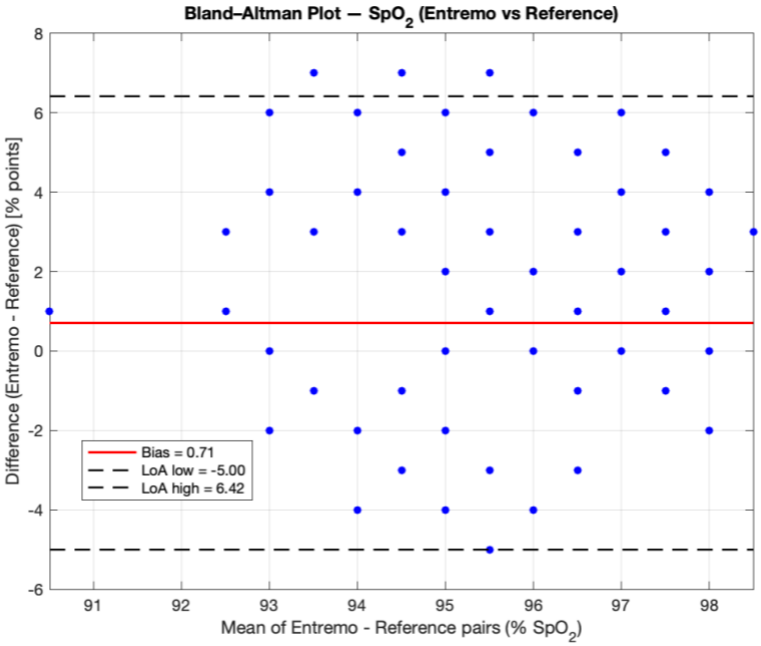


Figure 4.5: Bland-Altman analysis of measurements demonstrate the agreement level between the Entremo and reference devices.

4.2.3 Skin temperature accuracy

Similarly, as in the case of HR, reference measurement tests were conducted on 10 separate nights with the Entremo device and a certified medical thermometer, called Tucky [67]. Tucky is a patch that needs to be placed under the armpit, while the Entremo device sits on the wrist, therefore some variance between them is of course expected. Especially, when the arm stays outside of the blanket during sleeping, being exposed to cooler temperature than the rest of the body under the blanket. Additionally, skin temperature on the wrist is also by nature lower than at a location closer to the center of the body, so the goal of the calculation is to create a mapping that converts raw measurements to outputs that are as close to the reference device as possible. Filtering didn't improve the signal much as there were no sudden outliers in temperature sensing, so instead, two refinements were applied, a fixed and linear offset correction, to compare results with the reference values. The advantage of a constant +1.5°C offset to the signal is its simplicity and low resource consumption, however it doesn't respond well to ambient changes in the environment and yields a higher error rate. A more sophisticated version is the linear correction, which is determined by fitting an affine model $T_{corr} = aT_{raw} + b$ to paired Entremo – Tucky samples via ordinary least squares on a calibration subset, then applying the fixed coefficients to the remaining nights. Compared with a constant offset, a linear calibration can perform better, because it compensates both bias and night-to-night gain changes caused by ambient and contact variability, at negligible computational cost. It also provides diagnostics cues in the form of slope changes or intercept drift points indicating self-heating of the electronics. It also plays well with guard rules, so when SQI indicates poor contact, it is still easy to switch to a constant offset. **Figure 4.6** shows one of the night-long measurements including the reference device, the raw Entremo temperature reading, a constant offset and a mapping output using linear correction.

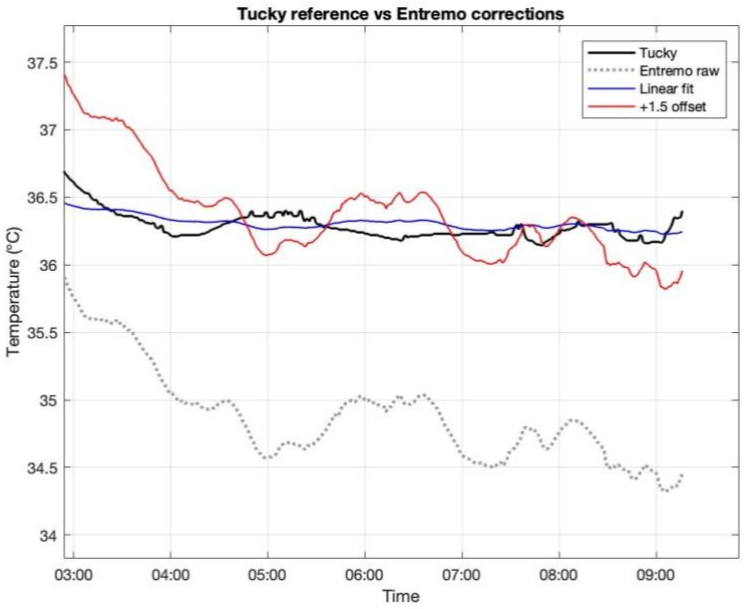


Figure 4.6: A night long temperature measurement with the Entremo prototype and Tucky reference device. The raw Entremo signal was corrected in two methods: a +1.5°C offset (red) and a linear correction model (blue).

Accuracy results of the 10 nights are shown in **Table 1**. The average MAE for the flat offset was 0.49°C and for the linear calibration it was 0.40°C. On seven nights, the linear model performed better, but three nights produced bigger outliers, in one case the error was more than 1°C. Possible reasons for the linear model failing could be poor contact or perfusion and cooler rooms than on calibration night.

Date	MAE_raw	Bias_raw	MAE_1.5	Bias_1.5	MAE_lin	Bias_lin
06.21	1.822 °C	-1.822 °C	0.764 °C	-0.322 °C	0.035 °C	-0.008 °C
06.10	1.371 °C	-1.371 °C	0.292 °C	0.129 °C	0.124 °C	-0.041 °C
06.03	1.378 °C	-1.345 °C	0.404 °C	0.155 °C	0.447 °C	0.267 °C
06.02	1.535 °C	-1.523 °C	0.245 °C	-0.023 °C	0.701 °C	0.514 °C
06.01	1.894 °C	-1.894 °C	0.463 °C	-0.394 °C	0.316 °C	0.246 °C
05.31	1.902 °C	-1.826 °C	0.537 °C	-0.326 °C	0.458 °C	0.339 °C
05.30	1.478 °C	-1.449 °C	0.525 °C	0.051 °C	0.375 °C	0.156 °C
05.27	1.175 °C	-0.72 °C	0.798 °C	0.78 °C	0.85 °C	0.647 °C
05.26	1.594 °C	-1.557 °C	0.31 °C	-0.057 °C	0.303 °C	0.188 °C
05.13	1.086 °C	-1.011 °C	0.588 °C	0.489 °C	0.441 °C	0.42 °C
Avg.	**1.524** °C	**-1.452** °C	**0.493** °C	**0.048** °C	**0.405** °C	**0.273** °C

Table 1: Mean Absolute Error (MAE) and Bias results during the 10 nights of temperature measurement for both the +1.5°C offset and linear correction model.

The flat +1.5°C offset performed robustly, never worse than 0.79°C MAE, but it is far from clinical grade accuracy. Across the ten nights, the average per-night Bland–Altman metrics for the linear-corrected temperature showed a bias of 0.27°C, SD of 0.62°C, and 95% limits of agreement of -0.94 to 1.49°C. Overall, neither methods met industry standard levels of accuracy, which require lower than 0.2°C error at temperatures below 37°C and 0.1°C error for temperatures above as set out in standard ASTM E1112-00(2016).

There were several limitations in this thesis project. Only a handful of test subjects were included in the study, because regulatory steps prevented the inclusion of any actual patients and institutions. As a result, no extensive dataset could be gathered to create a representative mapping model to account for different ages, skin types, differences in thermal regulation or exposure to different ambient environments. The use of contact gating was also absent from the experiment, there was no metric to signal skin contact pressure to filter out loose contact measurements.

4.3 Analysis summary

During evaluation the prototype met its primary feasibility goals while revealing clear priorities for the next iteration.

From a usability perspective, the project achieved a wireless, button-less bracelet with multi-day battery life, wipe-down sanitization, simple nurse-driven pairing and a seamless dashboard integration - features aligned with nurse, doctor and patient feedback summarized earlier.

On the technical side, feeding the data through an energy-efficient and robust processing pipeline – implementing baseline removal, bandpass filtering, peak detection, perfusion analysis, calibration models - produced comparable metrics across multiple use-cases, modalities and subjects (**Table 2**).

	N	Bias	SD	LoA min	LoA max	MAE	RMSE
HR	432	-0.09 bpm	3.9 bpm	-7.74 bpm	7.55 bpm	1.87 bpm	3.9 bpm
SpO_2	150	+0.71%	2.91%	-5.00%	+6.42%	2.31%	2.99%
Skin T_{lin}	882	+0.27 °C	0.62 °C	-0.94 °C	1.49 °C	0.40 °C	0.69 °C

Table 2: Summary of accuracy results for the three parameters measured by Entremo against reference devices.

5 Discussion and conclusion

In this final section of the report, the development results are discussed, conclusions are drawn, and the research questions are answered, which were the following:

Question 1: To what extent can a wrist-worn device reliably measure heart rate, oxygen saturation, and skin temperature – under typical ward conditions – compared with established monitoring methods?

Question 2: What design and workflow choices enable smooth, low-effort integration into everyday nursing routines while maintaining patient safety and hygiene?

5.1 Deliberation of a wireless wrist device

Wireless monitoring devices are not yet standard accessories in hospital wards, so the goal of this thesis was to present technical and design methods, which go into the development of a device that is intended to ultimately replace manual monitoring practices in healthcare institutions. The results of the development demonstrate that a wrist-based device is feasible in such an environment, though each parameter carries different levels of maturity. Heart rate accuracy met the ±3 bpm benchmark promising clinical use, with negligible bias against a certified chest-strap reference (**Figure 4.3**). Even though the wrist location yields weaker signal quality than a finger location, with optimal sensor placement and filtering, the signal is sufficient for peak detection, consistent with established consumer wearables. Oxygen saturation monitoring was also shown to be possible in motionless measurement periods, though current accuracy only narrowly satisfied the ISO requirement (RMSE $\leq$ 3%), reflecting the absence of a dedicated desaturation calibration study. Skin temperature monitoring produced stable results at trend level but did not reach the industry threshold for clinical-grade thermometry (**Table 1**). This result was expected, given the lack of certified commercial solutions even in the most advanced available devices. The temperature inconsistency at the wrist location is a significant hurdle in this wireless use-case, which requires more advanced solutions for calibration. In conclusion, these findings indicate that wrist-based sensing offers enough signal quality for early-warning and trend detection, but still cannot fully substitute established medical instruments operated manually, due to certain constraints and limitations. The Entremo prototype was validated only on a small group of healthy subjects, mostly under controlled conditions, so the findings cannot yet be generalized to diverse patient populations. SpO_2 relied on published calibration values instead of a dedicated desaturation study, and temperature measurement used rudimentary correction methods sensitive to ambient conditions. In addition, the processing pipeline lacked motion-compensation from accelerometers, limiting robustness during everyday activity.

5.2 Adoption in healthcare institutions

Beyond measurement performance, the study also validated a set of design choices that were critical for usability and adoption in healthcare institutions. Accuracy on its own is insufficient for acceptance in clinical wards; nurses, doctors, and patients all emphasized that workflow compatibility and trust are just as important. To this end, the Entremo prototype was deliberately designed as a wireless, screenless, button-free device that minimizes the chance of user error and patient tampering. A multi-day battery with magnetic charging was implemented to simplify charging routines, reduce cable clutter, and make device readiness visible at a glance. The hardware materials were chosen to allow quick wipe-down sanitization, ensuring safe re-use in fast-paced ward environments. These design choices were not arbitrary; they were derived directly from stakeholder feedback. Nurses emphasized effortless operation and reliable alerts, doctors prioritized remote access to continuous trends, and patients wanted comfort, familiarity, and discretion. Taken

together, the Entremo device addressed these needs with a streamlined, hospital-ready form factor, automatic pairing and data integration, and a familiar fitness-watch-like appearance that reduces patient anxiety (**Figure 4.1**). However, the study also highlighted the need for broader validation. Large-scale trials will be essential to demonstrate robustness under motion, skin variation, and diverse ward environments. Without such evidence, adoption will remain constrained to pilot use rather than wide institutional rollout.

5.2.1 Reflections

In addition to the technical evaluation, the Entremo prototype must be viewed in light of ethical, sustainability, and social considerations. Ethically, continuous monitoring produces highly sensitive patient data, which in real deployments must be safeguarded through encryption, GDPR compliance, and informed consent protocols. Sustainability was addressed in this project by opting for a reusable, wipe-down wristband rather than disposable sensors, thereby reducing medical waste, although the long-term challenge of battery recycling remains unresolved. Socially, the device has the potential to reduce nurse workload and enhance patient safety by automating routine monitoring tasks, yet its adoption depends on staff training, institutional trust, and integration with existing workflows. Patient comfort and familiarity with consumer wearables support acceptability, but there is also a risk of exacerbating inequalities if only better-resourced hospitals can afford such systems. These dimensions underline that technical innovation must be complemented by ethical safeguards, sustainable design, and socially responsible deployment strategies.

5.3 Future outlook

5.3.1 Desaturation study and temperature study

To advance beyond feasibility, a dedicated hypoxia study is necessary for establishing a representative R-to-SpO_2 mapping curve across the full 70–100% range. Similarly, temperature accuracy requires expanded datasets covering more subjects, skin types, and ambient conditions. Combining wrist readings with ambient reference sensors or multi-node models could substantially improve performance.

5.3.2 Noise reduction with accelerometer

A natural next step in improving signal quality is motion compensation through sensor fusion. Integrating accelerometer data into the processing pipeline would allow detection and removal of motion artifacts in real time.

5.3.3 Fall detection

While not implemented in the current prototype, fall detection emerged as a highly desired feature among elderly patients. Incorporating inertial sensing and threshold-based detection could position the device not only as a monitoring tool but also as a proactive safety system.

5.3.4 New measurement techniques

The Entremo device may also expand its parameter set. Blood pressure estimation via PPG, respiratory rate extraction, or hydration monitoring are areas actively researched in wearable sensing. Adding such capabilities, if validated, would enhance the clinical value of the device without compromising its compact design.

5.4 Closing Statement

In conclusion, this thesis demonstrated the feasibility of a wrist-worn, wireless vital sign monitoring device that can complement or eventually replace manual patient monitoring practices. The prototype achieved sufficient accuracy for heart rate, showed promising potential for SpO_2 and skin temperature, and validated key design choices grounded in stakeholder input. Limitations—particularly the absence of clinical calibration studies and small sample sizes—underscore the need for further work before more rigorous regulatory procedures can be started.

Nevertheless, the findings provide a strong foundation: the Entremo device shows how wireless wearables can ease nurse workload, improve early detection, and increase efficiency in non-intensive wards. While challenges remain, the project contributes a viable step toward transforming patient monitoring into a more continuous, automated, and patient-friendly process.

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